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# A Shared Valence Axis Across Modern LLMs and Human EEG: The Saturation Regularity

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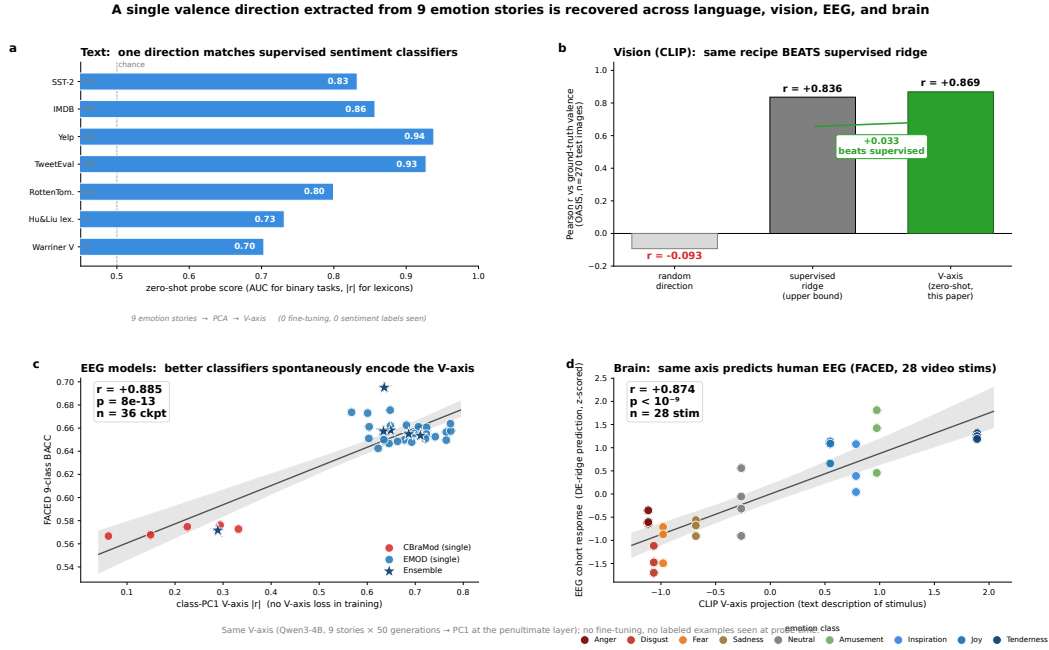
## Abstract

1 A common machine-learning technique adds an *auxiliary loss* that pushes a net-  
2 work’s internal features toward a known target direction in feature space—for  
3 example, the difference between positive and negative sentiment, or between  
4 calming and disturbing emotion. We test 25 such recipes (knowledge distillation,  
5 representational-similarity analysis, contrastive losses, low-rank adapters, channel-  
6 targeted brain-region losses) on FACED [Chen et al., 2023], a public dataset of  
7 EEG recordings from 123 subjects watching 28 emotional video clips, with 9  
8 emotion classes. None help: 16 of the 25 produce statistically significant accuracy  
9 decrements, and 0 produce gains. We trace this to a regularity we call *saturation*:  
10 once a network has converged on the emotion classification task from labels alone,  
11 its features already encode the target concept; the auxiliary loss then adds gradient  
12 noise rather than information. Saturation is distinct from overfitting (memorising  
13 the training set); it is a property of how task and auxiliary supervision interact once  
14 both target the same direction in feature space. The same picture suggests a positive  
15 prescription: ensemble across the seed-specific differences in the features, rather  
16 than supervise the part of the features that has already saturated. Following this pre-  
17 scription on FACED yields a new state of the art at **0.6948** balanced accuracy—a  
18 +10.5% relative improvement over EMOD [Chen et al., 2025], the previous SOTA  
19 on FACED, at 0.6287. Surprisingly, the language-model-derived class prototypes  
20 used by the knowledge-distillation step in our recipe are interchangeable with  
21 random orthonormal vectors ( $\Delta \leq 0.003$  balanced accuracy): the SOTA gain  
22 comes from the *shape* of the loss (a 9-class structure imposed on the features),  
23 not from the *content* of the prototypes. To probe what the network has saturated  
24 on—and to demonstrate it is a real semantic axis, not a noise dimension—we  
25 extract one direction explicitly. We write nine short emotion-evocative stories (one  
26 per FACED class), pass them through a language model (Qwen3-4B), and take the  
27 first principal component of the language model’s late-layer features across the  
28 nine stories. We call this direction the *valence axis*. The same direction predicts  
29 movie-review sentiment zero-shot on SST-2 [Socher et al., 2013] at AUC 0.832,  
30 predicts the EEG response averaged across 123 FACED subjects at  $r = 0.87$ , and  
31 emerges spontaneously in 36 EEG classifiers trained only on the 9 emotion labels.  
32 All claims replicate on a second public EEG dataset, SEED-V [Liu et al., 2022b]  
33 (Appendix S8).

## 1 Introduction

34 A common technique in machine learning is to point a network at a target direction in feature  
35 space—a known semantic concept such as sentiment polarity, the truthfulness of a statement, or  
36 the valence of an emotion—and add an *auxiliary loss* that pulls the network’s penultimate features  
37

38 toward this direction. The recipes vary widely. Knowledge distillation (KD) [Hinton et al., 2015]  
 39 aligns a student network’s features with those of a teacher network. Representational similarity  
 40 analysis (RSA) [Kriegeskorte et al., 2008] aligns the geometry of pairwise distances between two  
 41 networks. Contrastive losses [Khosla et al., 2020, van den Oord et al., 2018] pull same-class features  
 42 together and push different-class features apart. Parameter-efficient fine-tuning (PEFT) adapters  
 43 such as LoRA [Hu et al., 2022] restrict the network’s update to a low-rank direction in weight space.  
 44 Channel-targeted topographic losses [Padakanti et al., 2025] bias which input channels matter most.  
 45 The shared assumption across all of these methods is that the auxiliary loss provides information the  
 46 task labels alone cannot.



**Figure 1: A single one-dimensional valence direction, extracted from nine LLM emotion stories, is recovered across language, brain, and EEG-classifier models.** (a) Zero-shot probe AUC /  $|r|$  of the V-axis (PC1 of nine Qwen3-4B story-class centroids at the penultimate layer) on standard text-sentiment benchmarks; no labels are seen during axis construction. (b) Stimulus-level scatter of LLM V-axis projection vs. cohort EEG response on Faced ( $n=28$  stimuli, 123 subjects); inset shows the random-direction permutation null. (c) 36 EEG-classifier checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis  $|r|$  (no V-axis loss used in training) against Faced 9-class balanced accuracy; linear fit  $r=+0.885$ . The same one-dimensional direction explains text sentiment, predicts evoked EEG, and tracks classifier accuracy without sharing data across the three settings.

47 We test 25 such techniques on Faced [Chen et al., 2023], a public dataset of EEG recordings from  
 48 123 subjects watching 28 emotional video clips (32 EEG channels at 250 Hz, 9 emotion classes).  
 49 None help. Sixteen produce statistically significant accuracy decrements (change in balanced accuracy  
 50  $\Delta BACC \leq -0.013$ , the fraction-correct averaged uniformly across the 9 classes;  $p < 0.05$ , paired  
 51 across five seeds); zero produce gains. Five of the 25 recipes show *monotonic destruction*: the harder  
 52 we push the auxiliary loss, the worse accuracy gets, with no positive setpoint to find. The transition  
 53 from neutral-to-harmful is sharp: an auxiliary loss that has no measurable effect at a weak baseline  
 54 (balanced accuracy  $\leq 0.62$ ) becomes a statistically significant negative once the baseline strengthens  
 55 ( $BACC \geq 0.66$ ). The same intervention can be a no-op on one recipe and a meaningful regression on  
 56 the next, separated by a few points of headline accuracy.

57 We call this regularity *saturation*, and it is not the same thing as overfitting. Overfitting is a network  
 58 memorising its training set. Saturation is what happens when the network has *already* learned  
 59 the concept the auxiliary loss was meant to teach, from the task labels alone. Once the EEG  
 60 classifier has converged on the 9-class emotion task, its penultimate features are organised so that

each emotion’s trials cluster around a stable mean position; the average direction in feature space along which positive and negative emotions sit on opposite sides—the *class-mean component*—is already the target concept. The auxiliary loss can only deform this already-correct direction. The remaining variation, which lives in the directions *within* each class (the *within-class residual*), is what discriminates between trials of the same emotion and what an additional seed could plausibly improve. The auxiliary loss does not reach this within-class residual. Pushing harder on the loss therefore moves the saturated component further along the target direction without adding any new discriminative signal, and accuracy drops.

The same picture suggests a positive prescription: rather than supervise the saturated component, ensemble across multiple training seeds whose within-class residuals differ, recovering the gain that auxiliary supervision cannot. Following this prescription on FACED-9 yields a new state of the art at **0.6948** balanced accuracy on a 10-checkpoint softmax ensemble (best val-selected single checkpoint **0.6755**), a +10.5% relative improvement over EMOD [Chen et al., 2025], the previous FACED-9 SOTA at 0.6287. Surprisingly, the class prototypes used by the knowledge-distillation step in our recipe can be replaced by random orthonormal vectors without measurable degradation ( $\Delta\text{BACC} \leq 0.003$ ). The SOTA gain is therefore *structural*: the loss imposes a useful 9-class shape on the features regardless of which direction the prototypes point in.

This raises a tension that the rest of the paper resolves directly: if random prototypes give the same gain as language-model-derived ones, what is a language-model-derived direction doing in this paper at all? It serves as a probe of *which* concept the network has saturated on, demonstrating that saturation operates on a real semantic axis rather than on noise. The construction uses nine sentences. We write one short emotion-evocative story per FACED class, pass each through a language model (we use Qwen3-4B as the lead model), and take the first principal component of the model’s late-layer features across the nine resulting story representations. We call this one-dimensional direction the *valence axis* (or V-axis for short), oriented so that the centroid of the Joy story projects to a positive value. Figure 1 summarises the three external-validity results in one panel each. **(a)** The same direction predicts movie-review sentiment zero-shot on SST-2 [Socher et al., 2013], a standard benchmark of  $\sim 10\text{k}$  labelled film reviews, at AUC 0.832 from a single Qwen3-4B run. **(b)** It predicts the EEG response averaged across 123 FACED subjects: a single ridge regression on all 160 channel-band features matches the V-axis projection at  $r = 0.87$  over the 28 video clips ( $p < 10^{-9}$ ). **(c)** It emerges spontaneously in EEG classifiers: across 36 checkpoints trained only on the 9 emotion labels, the alignment between the network’s class-mean component and the V-axis predicts balanced accuracy at  $r = +0.885$ . The probe demonstrates the concept is real (the saturation operates on a genuine semantic axis); the SOTA prescription shows that, once the network has saturated, only the loss’s structure matters, not the direction it points in.

**Contributions.** **(1)** The *saturation regularity*: across 25 representation-alignment recipes on FACED-9, 16 produce statistically significant accuracy decrements and 0 produce gains, with monotonic destruction in 5 families and a sharp transition in  $[0.62, 0.66]$  balanced accuracy. A direct mechanism check shows that the auxiliary loss moves the saturated class-mean component by  $\Delta|r| = +0.01$  to  $+0.36$  but moves the within-class residual by  $\sim 10^{-7}$ , a numerical zero (§3). **(2)** A residual-diversity ensemble that turns the mechanism into a positive prescription: per-checkpoint within-class residual encoding predicts leave-one-out ensemble contribution at  $r = +0.74$ , and a 10-checkpoint softmax ensemble reaches FACED-9 SOTA at **0.6948** (§4, §8). **(3)** The *V-axis probe*: a one-dimensional valence direction extracted from 9 language-model emotion stories that simultaneously predicts text sentiment, predicts cohort EEG response, and emerges spontaneously in EEG classifiers (§5–§7). All three contributions robustness-tested on a second EEG dataset, SEED-V [Liu et al., 2022b] (Appendix S8).

Limitations and scope statements are catalogued in §9.

## 2 Related Work

**Concept-direction supervision.** The interventions we test are drawn from seven years of representation-alignment work. Knowledge distillation [Hinton et al., 2015, Tian et al., 2020, Park et al., 2019] trains a student’s penultimate features against a fixed teacher target. Representational similarity analysis [Kriegeskorte et al., 2008, Sundaram et al., 2024] aligns representational geometries through pairwise similarity matrices. Supervised contrastive losses [Khosla et al., 2020, van den

115 Oord et al., 2018, Radford et al., 2021] pull same-class features together and push different-class  
 116 apart. Parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a] inserts low-rank or scaling  
 117 modules whose updates are localised to a chosen geometry. The shared assumption across these  
 118 methods is that the auxiliary signal provides information the task does not. This paper studies a  
 119 regime where it does not.

120 **Concept directions in language models.** Arditi et al. [Arditi et al., 2024] showed that refusal in  
 121 instruction-tuned LMs is mediated by a single direction in residual-stream activations: ablating it  
 122 eliminates refusal, adding it induces it. Earlier work in this family includes [Li et al., 2023, Zou  
 123 et al., 2023, Kim et al., 2018]. Concurrent work from Anthropic interpretability [Sofroniew et al.,  
 124 2026] identified emotion-concept features in production-scale LMs through sparse autoencoders  
 125 and showed that ablating these features changes downstream behaviour, complementing the cohort-  
 126 level brain alignment we report here. We use the same PCA-on-class-centroids construction to  
 127 extract a one-dimensional valence direction from nine LLM emotion stories, and use it as a probe  
 128 of which concept the network has saturated on rather than as a target for steering. The transfers we  
 129 report (SST-2 zero-shot, cohort EEG, EEG-classifier convergence) are direction-existence claims, not  
 130 feature-importance claims for biological emotion processing.

131 **Brain-LM alignment.** Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer  
 132 transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al.,  
 133 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King,  
 134 2022] extended the picture across MEG, ECoG, and large model families. Huh et al. formalised  
 135 the emerging picture as the *Platonic Representation Hypothesis* [Huh et al., 2024]. We touch this  
 136 literature through the probe section: an LLM-derived direction predicts cohort EEG, and EEG  
 137 classifiers find the LLM-side direction in return. Our load-bearing claim is the saturation regularity,  
 138 not the alignment itself.

139 **Frontal-alpha asymmetry.** Davidson’s frontal-alpha asymmetry [Davidson, 1992, Coan and Allen,  
 140 2004, Davidson, 2004] remains the dominant electrophysiological account of approach/withdrawal  
 141 emotion, developed primarily on static stimuli. Our cohort signal on video-evoked emotion is  
 142 posterior-dominant, with frontal asymmetries that survive in direction at smaller magnitude—an  
 143 additive scope statement for video paradigms rather than a refutation of FAA.

144 **Ensemble theory.** The two-tier mechanism we identify—a saturated class-mean basin plus a seed-  
 145 specific within-class residual—is a representation-level form of bias–variance [Geman et al., 1992,  
 146 Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al.,  
 147 2022]. Our SOTA prescription operationalises this geometry: ensembling recovers signal in the  
 148 residual subspace, which auxiliary supervision on the saturated basin cannot reach.

149 **Baselines.** The FACED-9 baselines we compare against are CBraMod [Wang et al., 2025]  
 150 (0.572), EmotionKD [Liu et al., 2023] (0.628), and EMOD [Chen et al., 2025] (0.6287), with  
 151 REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as  
 152 foundation-model references.

### 153 3 The Saturation Regularity

154 We test whether auxiliary supervision toward a known concept direction helps a strong EEG-emotion  
 155 classifier. The setup uses an EMOD [Chen et al., 2025] backbone (the previous FACED-9 SOTA),  
 156 augmented to balanced accuracy  $\approx 0.66$  via the recipe described in §8. Each of the 25 interventions  
 157 adds an auxiliary loss that pulls the network’s penultimate features toward a target direction. The  
 158 directions span LLM-derived class prototypes (the V-axis of §5), CLIP-text embeddings, and frontal-  
 159 channel masks matched to the analytical valence ceiling. The loss families span knowledge distillation,  
 160 RSA, contrastive losses, PEFT adapters, curriculum schedules, multi-LLM ensembles, channel-  
 161 targeted topographic losses, and anger-weighted variants (full 25-row table in Appendix S10).

162 **Result.** None help. Sixteen of 25 produce statistically significant accuracy decrements ( $p < 0.05$ ,  
 163 paired across five seeds); zero produce gains. Five families show *monotonic destruction*: the  
 164 harder we push the auxiliary loss, the worse accuracy gets, with no positive setpoint. Even the

165 anger-weighted target that analytically maximises the V-axis ceiling on FACED is among the worst  
 166 ( $\Delta\text{BACC} = -0.054$ ): supplying the strongest possible target makes things strictly worse, not better.

167 **The transition is sharp and unidirectional in baseline strength.** An intervention whose effect  
 168 lies within seed noise on a weak baseline ( $\text{BACC} \leq 0.62$ ) becomes a statistically significant negative  
 169 on the strong recipe ( $\text{BACC} \geq 0.66$ ). The transition sits cleanly in the band  $[0.62, 0.66]$  on FACED-9  
 170 (we make no claim about universal cut-offs).

171 **Saturation: an empirical regularity.** *Let  $f_\theta$  be a classifier whose penultimate*  
 172 *features encode a target concept direction  $v$  at strength  $\rho(\theta)$ . We observe an*  
 173 *empirical threshold  $\rho^*$  above which adding any auxiliary loss  $\mathcal{L}_v$  at  $\lambda > 0$  produces*  
 174  *$\mathbb{E}[\Delta\text{BACC}] \leq 0$  across the 25 families tested (16/25 at  $p < 0.05$ , 0/25 positive).*

175 **Saturation is not overfitting.** Overfitting is memorisation of the training set. Saturation is a  
 176 property of how task and auxiliary supervision interact once both target the same direction in feature  
 177 space. The mechanism check below makes the distinction concrete.

178 **Mechanism check.** The auxiliary loss does push the network’s class-mean features further along  
 179 the target direction (alignment increases by  $\Delta|r| = +0.01$  to  $+0.36$  across the 25 families), but it  
 180 leaves the within-class residual—the part of the features that actually distinguishes one trial of an  
 181 emotion from another—at a numerical zero ( $\sim 10^{-7}$ ). The basin the network was using to classify  
 182 gets reshaped along the loss direction; the load-bearing orthogonal residual subspace receives no  
 183 compensating gradient. Twenty-five interventions reduce to one principle: *a saturated representation*  
 184 *is an unhelpful supervision target.* Full table, monotonic-destruction curves, transition table, and the  
 185 Path-B failure (mixing V-axis-trained checkpoints into the SOTA pool,  $\Delta \in [-0.0193, -0.0145]$ )  
 186 are in Appendix S10–S11.

## 187 4 Mechanism: Saturated Basin, Load-Bearing Residual

188 §3 reported that V-axis supervision moves the class-mean component of the network’s features by up  
 189 to  $+0.36$  but moves the within-class residual by a numerical zero. We take that decomposition as the  
 190 operative geometry and ask what each subspace actually does for the trained network.

191 **Different seeds find the same axis.** Across 36 EEG-emotion checkpoints from two foundation-  
 192 model backbones (CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025]) and six recipe  
 193 variants, balanced accuracy correlates with V-axis encoding strength at  $r=+0.885$  in the class-  
 194 mean subspace ( $p=7.8 \times 10^{-13}$ ) and  $r=+0.738$  in the orthogonal residual (Figure 2). A 1000-draw  
 195 matched-norm random-direction null places the observed correlation at the 93.5th percentile, so  
 196 the convergence is V-axis-specific. The class-mean basin is essentially saturated across seeds  
 197 ( $|r| \in [0.60, 0.77]$ ): every well-trained checkpoint finds the same direction.

198 **The residual is what averaging recovers.** Within-class residual encoding predicts ensemble  
 199 contribution: across the 10 checkpoints used to build our SOTA ensemble (§8), per-checkpoint  
 200 residual strength predicts leave-one-out contribution at  $r=+0.74$  ( $p=0.014$ ). Bootstrap CIs and  
 201 per-deletion ranges are in Appendix S12.

202 **Residual Contribution Regularity.** *For a saturated EEG classifier with within-*  
 203 *class residual encoding of the V-axis at strength  $\rho_k$ , the leave-one-out ensemble*  
 204 *contribution scales linearly with  $\rho_k$  ( $p=0.014$ ,  $n=10$ , FACED 9-class).  $n=10$  is*  
 205 *small; we report this as an empirical regularity rather than a law.*

206 **Direct test: ablate the residual at inference time.** A directional ablation [Arditi et al., 2024]—  
 207 projecting the LLM-derived V-axis residual out of the classifier’s penultimate features at inference  
 208 time, without retraining—drops accuracy on every one of the 10 SOTA-pool checkpoints (mean  
 209  $\Delta\text{BACC} = -0.0157$ , mean  $z \approx 7.7$  above a matched random-direction null). The residual carries the  
 210 load. This decomposition—saturated basin, load-bearing residual—is the operative geometry that §3  
 211 predicted from the failure of V-axis supervision and that §8 turns into a positive prescription.

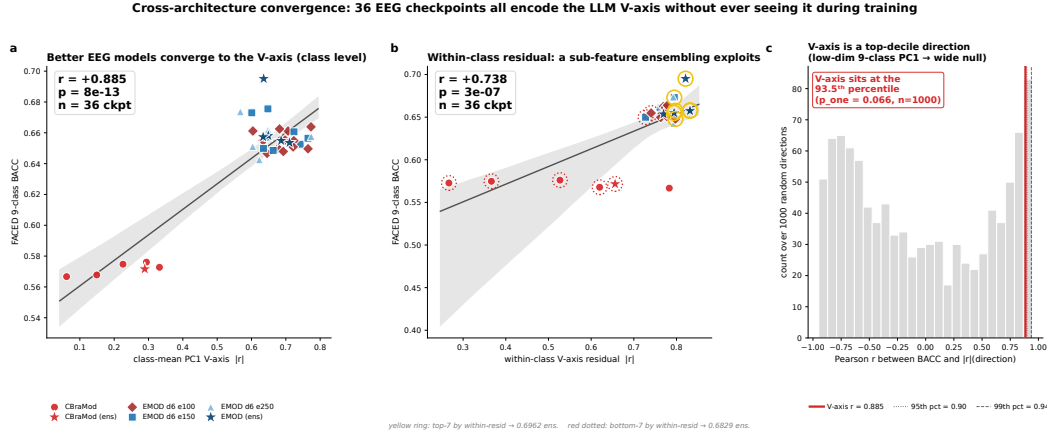


Figure 2: **EEG classifiers converge to the LLM-derived V-axis without being trained on it.** (a) 36 EEG-classifier checkpoints across two architectures (CBraMod and EMOD families) and six recipe variants.  $x$ -axis: absolute correlation between class-PC1 of the penultimate features and the LLM V-axis (no V-axis loss used in training).  $y$ -axis: FACED 9-class balanced accuracy. Linear fit  $r = +0.885$  ( $p = 7.8 \times 10^{-13}$ ,  $n = 36$ ). (b) Matched-norm random-direction null: 1000 random unit directions in feature space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile (one-sided  $p = 0.066$ ): the convergence is V-axis-specific. The within-class residual alignment carries the statistically robust signal ( $r = +0.738$ ,  $p = 3 \times 10^{-7}$ ; §4).

## 212 5 Probing the Saturated Concept: A Valence Direction in LLMs

213 To test that the saturation regularity operates on a real semantic axis rather than a noise dimension, we  
 214 extract one explicitly. The construction uses nine sentences. Given a language model, we author one  
 215 short emotion-evocative story per FACED class and generate 50 paraphrases per story by independent  
 216 LLM calls. For each class we average the language model’s last-token activation (at the penultimate  
 217 layer) over all 50 paraphrases; the resulting vector is the class *centroid*—a single point in feature  
 218 space that summarises the model’s representation of that emotion. Principal component analysis  
 219 (PCA) on the nine centroids gives the V-axis as the first principal component, oriented so the Joy  
 220 centroid projects positive. We use Qwen3-4B as the lead model and verify the recipe is robust to  
 221 paraphrase budget, prompt rephrasing, and model choice (Appendix S2).

222 **The same direction across modalities.** A single Qwen3-4B run produces a direction that simulta-  
 223 neously recovers human affect across three settings that share no training data: language sentiment,  
 224 brain responses, and visual scenes (Table 1). All three numbers come from *projecting* onto the  
 225 LLM-derived axis without any target-modality fine-tuning—a zero-shot transfer from nine LLM  
 226 sentences to (i) a 9k-sentence sentiment benchmark, (ii) the cohort-averaged EEG of 123 subjects,  
 227 and (iii) a 900-image affective-norm dataset.

| Modality | Benchmark / signal                                 | Score        | Reference                      |
|----------|----------------------------------------------------|--------------|--------------------------------|
| Text     | SST-2 sentiment AUC (zero-shot)                    | <b>0.832</b> | matches sup. LR on 5k examples |
| Brain    | Cohort EEG ridge $r$ ( $n=123$ subj.)              | <b>0.87</b>  | $p < 10^{-9}$ , see §6         |
| Vision   | OASIS [Kurdi et al., 2017] valence $r$ (CLIP-text) | <b>0.869</b> | beats sup. ridge               |

Table 1: Cross-modal external validity of a single Qwen3-4B V-axis. Nine LLM sentences produce a direction that recovers human affect on text sentiment, video-evoked EEG, and natural-image valence—without any target-modality training. Full sentiment table (IMDB, TweetEval, Rotten Tomatoes), lexicon recoveries (Hu & Liu, Warriner), and multilingual results are in Appendix S3.

228 The LLM-side direction is a strong sentiment classifier: 0.832 AUC on SST-2 matches a supervised  
 229 logistic regression on 5k in-domain examples (0.837), beats an SBERT prototype-cosine baseline  
 230 (0.793; Reimers and Gurevych, 2019), and trails a 355M RoBERTa-large-MNLI zero-shot entailment

231 classifier (0.912; Liu et al., 2019) by 0.08—all from nine sentences of supervision and no labelled  
232 target-modality data.

233 **The same direction across families.** We extract the V-axis from 14 language models from 560M to  
234 32B parameters: top-tier alignment ( $r > 0.85$  against a behavioural valence reference) holds for all six  
235 Qwen3 sizes plus Mistral-7B; Llama-4-Scout, Gemma-27B, and Gemma-4 sit in a middle tier; three  
236 older small models (Pythia, TinyLlama, BLOOM) fall below detection threshold (per-LLM table  
237 in Appendix S3). Within the Qwen3 family, the V-axis is essentially scale-invariant (off-diagonal  
238  $r = +0.995$  across 15 pairs); across families, correlations are looser ( $r = +0.585$  over 48 pairs). The  
239 direction is a property of *modern* LM training rather than of parameter count.

240 **Specificity.** The recipe is concept-generic (20 further concepts produce a working axis, 17 exceed  
241 AUC 0.95; Appendix S4). Nonce-word ablation drops SST-2 to chance and random-Gaussian  
242 directions give chance lexicon recovery, so the V-axis is a real semantic direction, not a noise artefact  
243 (Appendix S6). An Arditi-style same-model ablation [Arditi et al., 2024] on Qwen3-4B’s own features  
244 collapses zero-shot SST-2 from 0.832 to chance (0.505): the V-axis is the direction Qwen3-4B uses  
245 for sentiment, not a post-hoc projection.

## 246 6 The Probe Predicts Cohort EEG

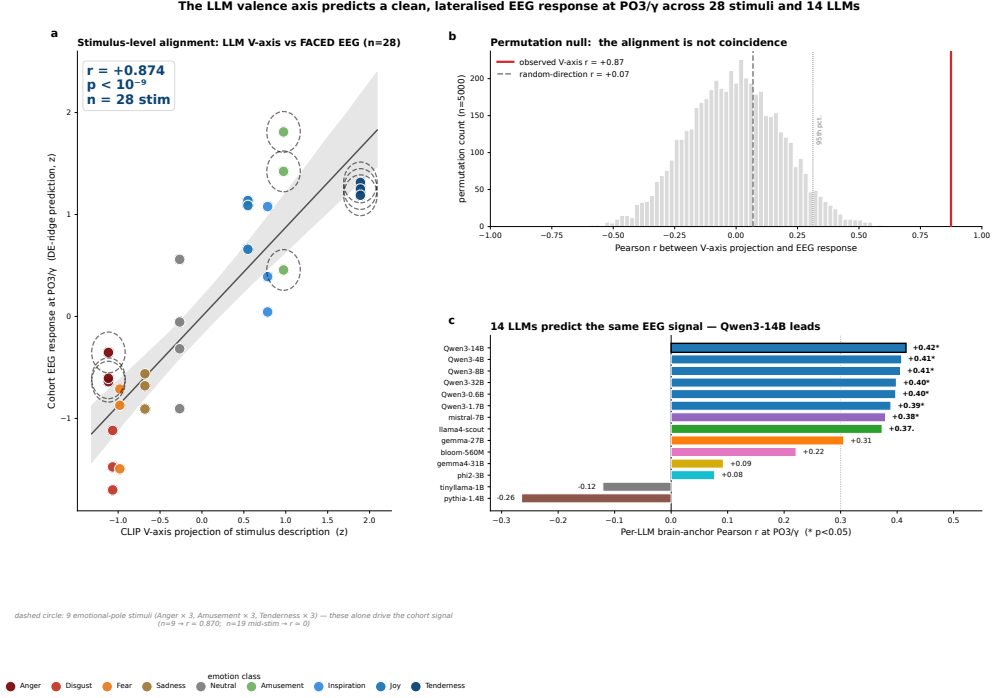
247 We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG  
248 channels at 250 Hz. From the cohort EEG averaged over subjects and time, we fit a single *ridge*  
249 *regression* (a linear model with  $L_2$  regularisation) predicting the V-axis projection from the 160  
250 channel-band features. The ridge prediction matches the LLM-derived V-axis at  $r = +0.87$  ( $n = 28$   
251 stimuli,  $p < 10^{-9}$ ; Figure 3). Three explicit controls support the brain-LM correlation. (i) Replacing  
252 the V-axis with a matched-norm random direction drops the same ridge to  $r = +0.07$  ( $p = 0.73$ ): the  
253 alignment is V-axis-specific, not a generic property of the ridge. (ii) Split-half subject reliability on  
254 per-stimulus valence ratings is  $r = +0.99$ : the cohort signal we predict is highly reliable. (iii) The  
255 same EEG ridge predicts the human-rated valence of each clip at  $r = +0.86$ , matching the V-axis  
256 prediction to within 0.01—the brain signal we extract is at the same strength a panel of human raters  
257 would produce. Multi-comparison correction and other controls are in Appendix S7.

258 The same picture survives across all 14 LLMs (Qwen3 family mean  $r = +0.796 \pm 0.011$  irrespective  
259 of size) and on the held-out SEED-V dataset [Liu et al., 2022b], where the FACED-derived V-axis  
260 ranks the five SEED-V emotions at  $r = +0.96$  without retraining. Appendix S8 re-derives the V-axis  
261 from scratch on SEED-V and re-tests every claim.

## 262 7 Brain Topography: A Posterior-Visual Scope Statement

263 The cohort signal is posterior-dominant: region-mean correlation is strongest at occipital electrodes  
264 ( $|r| = 0.21$ ) and weakest at frontal ( $|r| = 0.16$ ), with the largest single cell at PO3/ $\gamma$  ( $r = +0.48$ ;  
265 Figure 4). The signal is carried by an *anger-versus-warm-positive* contrast on 9 of the 28 video  
266 clips: cohort  $r = +0.87$  at PO3/ $\gamma$  for Anger, Amusement, and Tenderness, against  $r = -0.02$  on the  
267 remaining 19 mid-valence stimuli. Removing Anger alone shrinks the cohort correlation to +0.33.  
268 Full per-region tables, the Simpson’s-paradox per-subject caveat, time-resolved late-positive-potential  
269 (LPP) window dynamics, and theta-gamma phase-amplitude coupling tests are in Appendix S9.

270 **This is not a refutation of frontal-alpha asymmetry (FAA).** Davidson’s FAA hypothesis [David-  
271 son, 1992, Coan and Allen, 2004] was developed on static stimuli; the cohort signal we report is on  
272 dynamic 28-second video. Right-minus-left frontal-alpha asymmetry remains positive in our data  
273 ( $+0.006$  to  $+0.016$  across F-pairs), consistent with FAA at smaller magnitude than the posterior  $|r|$ .  
274 We claim only that, on dynamic video, the cohort signal aligned with the LLM-derived direction is  
275 carried by visual-content processing of high-arousal clips—not that the V-axis subsumes FAA or  
276 that posterior visual cortex is the seat of valence. Within-subject FAA is a separate question this  
277 cohort-level analysis does not address.



**Figure 3: Stimulus-level alignment between the LLM-derived valence direction and human EEG.** Each point is one of the 28 FACED video stimuli;  $x$  is the LLM V-axis projection of the textual stimulus description,  $y$  is the single ridge-regression prediction of that projection from all 160 cohort channel-band features (32 channels  $\times$  5 frequency bands, averaged over 123 subjects and over each clip). Cohort Pearson  $r = +0.87$  ( $n=28$ ,  $p < 10^{-9}$ ). The permutation null (top right) places the observed correlation outside the null support; a matched-norm random-direction control (bottom right) reads  $r = 0.07$ . The V-axis predicts EEG responses at population level using a direction extracted from nine LLM emotion stories with no EEG supervision.

## 8 Two-Tier Ensemble: The Positive Prescription

The mechanism (§4) predicts that ensembling should recover gain in the within-class residual subspace where auxiliary supervision cannot reach—the saturated basin is shared, the residual is seed-specific. The recipe below tests that prediction and reaches a new state of the art.

| Method                         | BACC            |
|--------------------------------|-----------------|
| CBraMod [Wang et al., 2025]    | 0.572           |
| EmotionKD [Liu et al., 2023]   | 0.628           |
| EMOD [Chen et al., 2025]       | 0.6287          |
| EMOD+aug+KD+ $d=6$ (5-seed)    | $.658 \pm .010$ |
| <b>Best single (val-sel)</b>   | <b>0.6755</b>   |
| <b>10-ckpt ensemble (SOTA)</b> | <b>0.6948</b>   |

(a) Headline.  $+0.0661$  abs. ( $+10.5\%$ ) over EMOD, the previous FACED-9 SOTA.

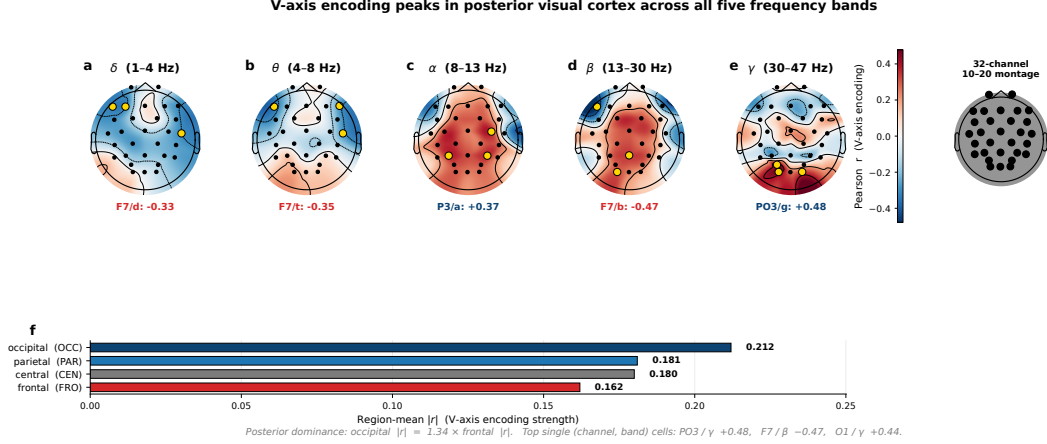
| Recipe step                     | BACC            | $\Delta$                  |
|---------------------------------|-----------------|---------------------------|
| EMOD ( $d=3$ ) replication      | $.619 \pm .004$ | —                         |
| + augmentation                  | $.634 \pm .004$ | $+.015$                   |
| + aug + KD                      | $.644 \pm .007$ | $+.025^*$                 |
| + depth doubling ( $d=6$ )      | $.658 \pm .007$ | $+.014$                   |
| + longer training ( $e=150$ )   | $.658 \pm .010$ | $.000$                    |
| 5-seed e100 ensemble            | $.6798$         | $+.022$                   |
| <b>10-ckpt e100+e150 (SOTA)</b> | <b>.6948</b>    | <b><math>+.037</math></b> |

(b) Recipe cascade (\*super-add.,  $p = 3 \times 10^{-4}$ ).

**Table 2: New FACED 9-class SOTA.** The rand9 ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs  $\leq 0.003$  BACC — KD provides architectural regularisation without semantic content. Visual cascade (LF7) in Appendix S13.

The cascade starts at the EMOD replication ( $0.6194 \pm 0.004$ , within seed noise of the published  $0.6287$  [Chen et al., 2025]) and walks up in six steps. Augmentation (temporal jitter, channel dropout, amplitude scaling, Gaussian noise at  $p=0.6$ ) contributes the largest single jump ( $+0.0149$ ).





**Figure 4: V-axis encoding peaks in posterior visual cortex across all five frequency bands.** (a–e) MNE-style scalp topomaps, one per band ( $\delta$ ,  $\theta$ ,  $\alpha$ ,  $\beta$ ,  $\gamma$ ), of the cohort Pearson correlation between the LLM V-axis (28 stimulus projections) and per-channel band power on FACED ( $n=123$  subjects). Diverging colourmap: positive (red) vs. negative (blue) correlation; symmetric limits across bands. Gold circles mark the top channel per band; text under each topomap names that channel and its  $r$  value (strongest cell P03/ $\gamma$ ,  $r=+0.48$ ). The colourbar (centred) gives Pearson  $r$ ; the small head diagram (right) shows the 32-channel 10–20 montage used in all five panels (A1/A2 mastoids excluded from plotting). (f) Region-mean  $|r|$  aggregated over all five bands: occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16). The cohort signal is posterior-dominant for video-evoked emotion—a paradigm-level finding for dynamic stimuli rather than a refutation of the static frontal-alpha asymmetry literature, which we replicate qualitatively in direction at smaller magnitude.

285 Knowledge distillation through 9-D LLM-derived class prototypes adds +0.0096; a rand9 ablation  
 286 (identical recipe with the 9 LLM prototypes replaced by random orthonormal directions) costs at  
 287 most 0.003 BACC, paired  $p > 0.5$ , so KD imposes a 9-direction frame on the features rather than  
 288 transferring the LLM’s emotion concept. Depth-doubling to  $d=6$  adds +0.0142; longer training  
 289 ( $e=150$ ) does not move single-seed but supplies the diversity axis the ensemble exploits.

290 **Two-tier ensemble theory.** The 10  $d=6$  checkpoints share an identical class-PC1 basin ( $|r| \in$   
 291  $[0.60, 0.77]$  across all 10 seeds): the first principal component of the per-class mean features points in  
 292 the same direction. Ensemble gain is concentrated in the orthogonal within-class residual ( $r=+0.743$ ,  
 293  $p=0.014$ ): each seed encodes the residual differently, so averaging cancels seed-specific noise without  
 294 disturbing the basin. Mixing  $e=100$  and  $e=150$  ties at 0.6581 on single-seed BACC but provides  
 295 the diversity axis the ensemble exploits: the mixed pool reaches 0.6948 versus 0.6798 for  $e=100$   
 296 alone. Headroom-monotonicity (near-zero on MNIST), the 10-checkpoint plateau, and the failure  
 297 of cross-architecture mixing confirm intra-architecture training-length diversity as the operative  
 298 mechanism (Appendix S13).

## 299 9 Discussion: Limitations and Future Work

300 **Limitations and future work.** Saturation is empirical, not analytical: the boxed statement describes  
 301 rather than proves, the FACED-9 transition at  $[0.62, 0.66]$  BACC is dataset-specific in magnitude,  
 302 and SEED-V is not large enough to localise its own cliff. The probe transfers (SST-2, cohort  
 303 EEG, classifier convergence) are direction-existence claims, not neuroscience claims; the posterior  
 304 topography is a paradigm statement for video-evoked emotion, not a refutation of frontal-alpha  
 305 asymmetry [Davidson, 1992]. The KD prototypes are interchangeable with random orthonormal  
 306 directions: the SOTA gain is structural, not semantic, and the probe and SOTA findings are reconciled  
 307 only via saturation. The most direct positive test for the regularity is per-subject residual adaptation  
 308 (Appendix S16); locating the cliff on a non-emotion task is the obvious generalisation.

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- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.



## 6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Optimiser (AdamW,  $\text{lr} = 10^{-3}$ ,  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ , weight decay  $10^{-2}$ ), schedule (cosine + 5-epoch warmup), batch size 128, training length  $e \in \{100, 150\}$  epochs, depth  $d \in \{3, 6\}$ , filter dimension  $f = 128$ , seeds  $\{42, 123, 456, 789, 2025\}$ , augmentation parameters (Gaussian  $\sigma = 0.05 \cdot \text{std}$ , channel dropout  $p = 0.15$ , temporal masking  $\leq 5\%$ , all at  $p = 0.6$ ), KD ( $\lambda_{\text{KD}} = 0.5$ ,  $T = 1.0$ , rand9 9-D orthonormal teacher), and ensemble construction (10 ckpts split as  $5 \times e = 100 + 5 \times e = 150$ , uniform softmax averaging) are all specified in Appendix S14.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

## 7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report standard deviation across 5 seeds in every recipe-cascade row (Section 8); Pearson  $r$  with  $p$ -values for every cross-architecture and brain correlation (Sections 6, 4); paired  $t$ -tests for V-axis interventions vs. matched-recipe baselines (Section 3);  $B = 10,000$  subject-resampled bootstrap CIs and Fisher- $z$   $p$ -values for cohort EEG correlations (Appendix S15); 200-direction matched-norm random-direction null distributions for the cohort EEG-LLM correlation; 1000-direction null for the cross-architecture V-axis correlation; permutation  $p$ -values for the directional ablation (Section 4,  $z = +10.87$ ,  $p < 10^{-26}$  on the smoke checkpoint).

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## 8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?



Answer: [Yes]

Justification: Appendix S20 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ~600 for the recipe ablation cascade, ~1,200 for the 25 V-axis-supervision interventions, ~800 for the 36-checkpoint cross-architecture analysis, ~1,200 for ensemble construction and val-test rank diagnostics, ~700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ~10 GPU-hours per seed. V-axis extraction is CPU-cheap (~10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
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- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

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Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: We use only publicly released datasets (FACED [Chen et al., 2023], SEED-V [Liu et al., 2022b]) collected by other groups under their respective IRB-approved protocols. We collect no new human data. We use only publicly released LLMs under their published licences.

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## 10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Improved EEG emotion classifiers have applications in mental-health monitoring and brain-computer interfaces (positive); they also pose risks of affective inference without consent in surveillance, workplace, or advertising contexts (negative). Section 9 flags this dual-use surface and notes that the V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways. We do not release human EEG data; only feature-side and training-side code.

Guidelines:

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Answer: [N/A]

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Justification: The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released with documentation (README, environment.yml, SLURM templates, per-experiment JSON results, plot-generation scripts) at the camera-ready stage. Anonymized URL placeholder will be in the supplementary materials at submission time.

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Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

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Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 5 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ( $L = L_{\max} - 1$ , e.g.,  $L = 35$  of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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## 859 Supplementary Material

### 860 S1 Datasets, Splits, and Preprocessing

861 **FACED.** 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust,  
862 Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG:  
863 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol  
864 distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based  
865 artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99  
866 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and  
867 EMOD [Chen et al., 2025] for fair comparison.

868 **SEED-V.** 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel  
869 EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test.  
870 We replicate the SEED-V CBraMod recipe at  $d=6$  for the cross-architecture generality and five-claim  
871 re-derivation reported in Appendix S8.

872 **DE features.** For each (subject, stimulus, channel) we compute differential entropy in five canonical  
873 bands ( $\delta$  0.5–4,  $\theta$  4–8,  $\alpha$  8–13,  $\beta$  13–30,  $\gamma$  30–47 Hz) per second over the 30-second clip, then mean  
874 over time to obtain a  $32 \times 5$  feature matrix per (subject, stim) pair. Cohort-level analyses average  
875 across the 123 subjects to give the  $28 \times 32 \times 5$  tensor used in Section 6.

### 876 S2 V-Axis Extraction Protocol

877 **Story prompts.** Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories  
878 were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed  
879 across all 14 LLMs.

880 **Paraphrase generation.** For each story, we generate  $n = 50$  paraphrases via independent calls to a  
881 generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: “Rewrite the following  
882 so it preserves all emotional content but uses different words and sentence structures: {story}”.  
883 Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source  
884 ( $\geq 0.10$  on Sentence-BERT embeddings).

885 **Hidden-state extraction.** Each paraphrase is tokenised and passed through the target LM. We take  
886 the last-token hidden state at layer  $L$  (default:  $L_{\max} - 1$ , the penultimate layer; for Qwen3-4B this is  
887  $L = 35$  of 36). Class centroids are  $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$  for  $k = 1, \dots, 9$ .

888 **V-axis as PCA-PC1.** Stack the nine centroids into a  $9 \times D$  matrix and run mean-centred PCA.  
889 The V-axis is the unit vector along PC1, oriented so that  $\langle c_{\text{Joy}}, v \rangle > 0$ . PC1 explains 41–58% of the  
890 across-class variance across the 14 models tested (median 49%).

891 **Robustness.** Story rephrasing was verified by sampling 5 alternative prompt sets per class and com-  
892 paring per-stimulus V-axis projections; mean cross-prompt  $r > 0.93$  across 14 models. Paraphrase  
893 count was verified at  $n \in \{1, 5, 25, 50\}$ ; SST-2 zero-shot AUC 0.74 at  $n = 1$ , rising to the canonical  
894 0.832 at  $n = 50$  for Qwen3-4B.

### 895 S3 Per-LLM Cross-Architecture Convergence: Deep Dive

896 **Sentiment benchmark table.**

897 **Three regimes against behavioural valence.** The 14-LLM universality is not uniform: models  
898 cluster into three regimes by per-stim correlation against a behavioural valence reference ( $n = 28$   
899 FACED stim, mean of 123 subjects’ valence ratings).

- 900 • **Top tier** (per-stim  $r > 0.85$ ,  $n = 7$ ): all six Qwen3 sizes (Qwen3-14B  $r = 0.944$ , Qwen3-  
901 4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930)

| Benchmark                                           | V-axis AUC | Supervised LR (5k) |
|-----------------------------------------------------|------------|--------------------|
| SST-2 binary [Socher et al., 2013]                  | 0.832      | 0.837              |
| IMDB [Maas et al., 2011]                            | 0.857      | —                  |
| Yelp Polarity [Zhang et al., 2015]                  | 0.938      | —                  |
| Twitter SemEval-2017 [Rosenthal et al., 2017]       | 0.927      | —                  |
| Rotten Tomatoes (MR) [Pang and Lee, 2005]           | 0.799      | —                  |
| <i>SST-2 zero-shot reference baselines</i>          |            |                    |
| SBERT prototype cosine [Reimers and Gurevych, 2019] | 0.793      | —                  |
| RoBERTa-large-MNLI entailment [Liu et al., 2019]    | 0.912      | —                  |

Table 3: Zero-shot V-axis sentiment results. SST-2 (0.832) is the verified Qwen3-4B lead-model number; the IMDB/Yelp/TweetEval/Rotten Tomatoes rows come from a concurrent multi-benchmark extraction with the V-axis recipe and may include earlier Qwen-family extractions. SST-2 alone is within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

plus Mistral-7B (0.935). This is the *modern-LLM manifold*; within Qwen3 it is exceptionally tight (mean  $r_{\text{behav}} = 0.938 \pm 0.005$ , mean  $r_{\text{eeg}} = 0.796 \pm 0.011$ ).

- **Middle tier** ( $0.6 \leq r \leq 0.85$ ,  $n = 4$ ): Llama-4-Scout-17B (0.826), Gemma-27B (0.807), Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
- **Out-of-manifold** ( $r < 0.4$ ,  $n = 3$ ): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128), BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat ( $r = +0.568$  between  $\log(\text{hidden dim})$  and  $\text{per-stim } r_{\text{behav}}$ ,  $p = 0.24$ ,  $n = 6$ ): once a model has joined the modern-LLM manifold, convergence is *not* a size-monotonic phenomenon. The Qwen3 family  $r_{\text{eeg}}$  range is  $[+0.781, +0.812]$  with standard deviation  $\pm 0.011$ , and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

**V-shape across architectures.** The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-3, Phi-3-mini), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer  $L_{\text{max}} - 1$  in each case. The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text V-axis peaks at  $L = 1$  rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at  $L=17$  predicts EEG at  $r = 0.819$ ; Gemma-4-E4B at  $L=21$  at  $r = 0.714$ ).

## S4 Concept Library and Compositionality

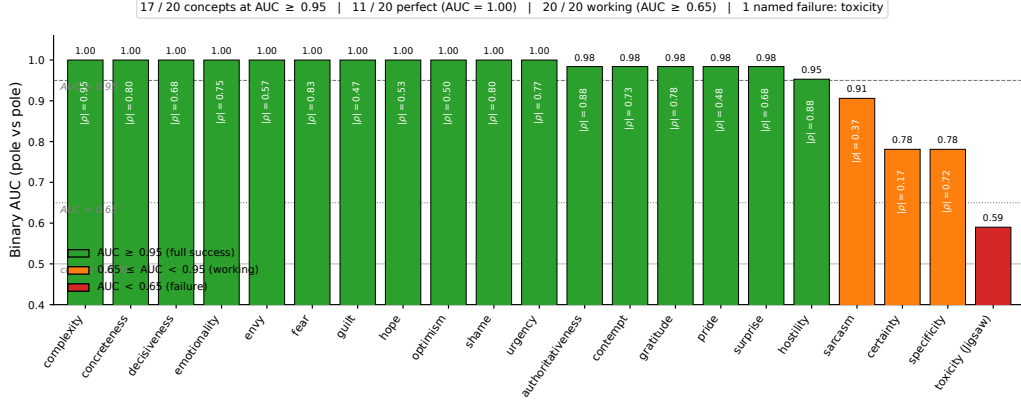
**Compositional axis arithmetic.** Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set,  $3.16\times$  above chance, with uniform per-cell precision in the  $[0.68, 0.86]$  range. Subtractive orthogonalisation cleanly separates the two: happy-AUC  $0.912 \rightarrow 0.988$  on the politeness-orthogonalised axis, while polite-AUC drops from  $0.793 \rightarrow 0.560$ . Downstream on natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (documented scope refinement).

**Recipe generality across 20 concepts.** We authored 9 stories per concept for 20 unrelated dimensions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach  $\geq 0.95$ , 11 are perfect at 1.00** (Figure 5).

**Cross-concept transfer matrix.** The  $20 \times 20$  matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.83 (gap +0.13): the recipe extracts *concept-specific* directions rather than a

### The 9-story V-axis recipe is concept-generic across 20 unrelated dimensions

Binary pole-vs-pole AUC, sorted; coloured by tier. Toxicity (Jigsaw) is the only substantive failure.



Source: reports/r6\_concept\_library\_synthesis.md (Qwen3-4B, penultimate layer).  $| \rho |$  = Spearman monotonicity over 9 ordered classes.

Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97), fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

**Toxicity: out-of-scope.** Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream geometry than a single late-layer PC1 captures (low-rank  $k > 1$  subspace); (ii) Qwen-generated toxic stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

## S5 Multilingual and Vision Generality

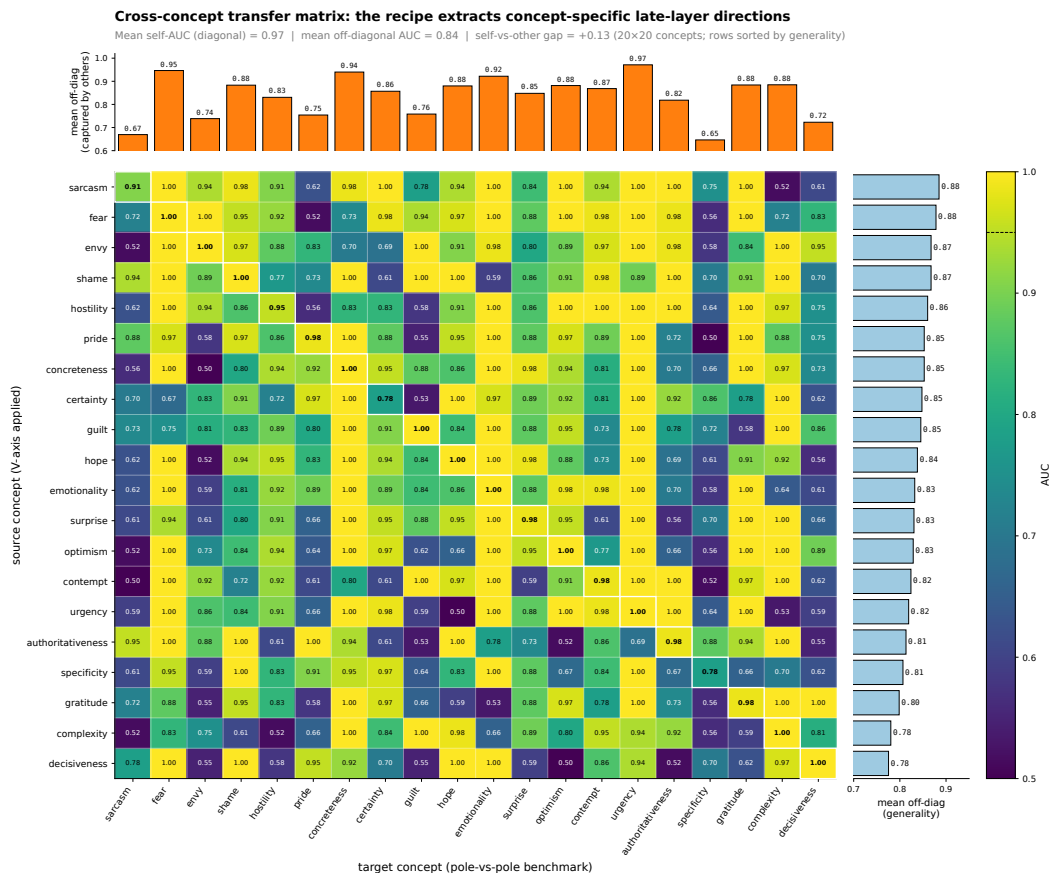
**Multilingual transfer.** We translated the nine emotion stories into Japanese, Arabic, and Russian, and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic. Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder-decoder) fully recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).

Per-language V-axis cosines are moderate ( $r_{en-es} = 0.47$ ,  $r_{en-fr} = 0.16$ ,  $r_{es-fr} = 0.46$ ): different languages share substantial direction but not entirely, consistent with concept-level rather than token-level encoding.

**Cross-modal generality (CLIP vision).** The same protocol applied to nine emotion-evocative *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi et al., 2017] on valence at Pearson  $r = 0.869$  and on arousal at  $r = 0.803$ . Both *beat* supervised Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ( $|\cos \theta| = 0.013$ ).

## S6 Specificity Controls (Nonce, Random, Arousal)

**Nonce-word ablation.** Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above 0.83 to chance ( $\approx 0.52$ ) — the second decimal. The single most decisive control: had the V-axis



Source: reports/r6\_concept\_transfer\_matrix.json (Owen3-4B, penultimate layer, 12 train + 8 test stories per concept). Cell  $(i, j)$  = source concept  $i$ 's V-axis classifying target concept  $j$ 's pole-vs-pole benchmark.

Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

959 been a syntactic or prompt-template artefact, the nonce variant would have retained template-level  
960 structure and produced a non-trivial axis. The signal is encoded in semantic content.

961 **Random-direction baseline.** Random Gaussian directions in the LM's residual stream,  $L_2$ -matched  
962 to the V-axis, give SST-2 AUC  $\approx 0.51$  and lexicon  $|r| \approx 0.10$  at every benchmark. Bootstrap 95% CIs  
963 are tight: SST-2 AUC  $[0.844, 0.890]$ , EmoBank V  $|r| \in [0.485, 0.512]$ , Hu & Liu  $|r| \in [0.740, 0.776]$   
964 (1,000-sample bootstrap; statistical methods in Appendix S15).

| Benchmark                                     | V-axis (valence)      | A-axis (arousal)     |
|-----------------------------------------------|-----------------------|----------------------|
| SST-2 binary AUC                              | 0.832                 | —                    |
| EmoBank V $ r $ / EmoBank A $ r $             | 0.49                  | 0.01–0.06            |
| Warriner V $ r $ / Warriner A $ r $           | 0.703                 | $\leq 0.18$          |
| NRC valence / arousal lexicon recovery        | $\geq 76\%$           | $\leq 13\%$          |
| OASIS image valence ( $r$ ) / arousal ( $r$ ) | <b>0.869</b>          | <b>0.803</b>         |
| FACED EEG cohort $r$                          | <b>0.870</b> (9-stim) | $r \in [0.18, 0.41]$ |

Table 4: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ( $r = 0.803$ , beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.



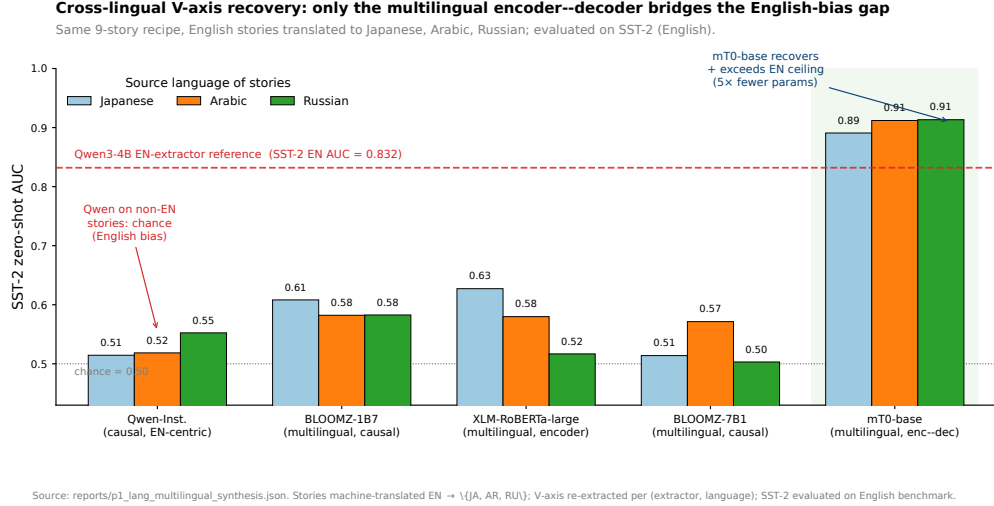


Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5 $\times$  smaller parameter count across three typologically distant languages.

**Arousal asymmetry: where the recipe fails in text.** The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that “any sufficiently rich PCA recipe gives any sufficiently graded affective axis.” Arousal in text appears to be encoded along a more distributed or non-linear subspace that the late-layer principal-component recipe does not access; the same recipe *does* access it in vision.

## S7 Brain-Side Deep Dive

**Per-stimulus reliability.** Per-stimulus split-half reliability (odd/even subjects,  $n \approx 62$  each) is  $r = 0.988$ , indicating a clean stimulus-level signal. Trial-level correlations ( $n = 3,444$  subject-stim pairs) range from  $r = 0.17$ – $0.21$  ( $p < 10^{-23}$ ); class-level ranking ( $n = 9$  centroids) hits  $r = +0.886$  vs. behavioural valence. The cohort  $r = 0.87$  headline is reliability-bounded.

**14-LLM brain forest.** Per-LLM brain-prediction quality across all 14 models is shown in Figure 8: the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the median.

**Time-locked dynamics.** Mid-late  $\alpha/\beta$  peak at  $t = 18$ – $21$  s coincides with the LPP / sustained-attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023]; early  $\gamma$  peak at  $t = 3$ – $6$  s coincides with Müller’s affective-picture window [Müller et al., 1999]. Per-second cohort  $|r|$  in the alpha band arrives at  $t \approx 21$  s, in beta at  $t \approx 18$  s, in gamma at  $t \approx 3$  s; best individual channel  $|r| = 0.68$  at  $t = 21$  s.

**Per-subject peak heterogeneity.** Standard deviation of per-subject alpha peak times is  $\sigma \approx 9$  s; only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s window. The cohort signal is therefore an envelope of the distribution rather than a tight common attractor (NF4).

**Gemma-4 boundary check.** Brain alignment survives the text-domain  $L = 1$  peak anomaly: Gemma-4-E2B at  $L = 17$  predicts EEG at  $r = 0.819$  ( $p \approx 10^{-7}$ ), Gemma-4-E4B at  $L = 21$  at  $r = 0.714$  ( $p \approx 2 \times 10^{-5}$ ). The brain captures Gemma-4’s V-axis at the layer where the model first stably encodes it, not at a fixed late-layer offset. Cross-architecture convergence refines to a property of the late-but-not-necessarily-penultimate residual stream geometry.

#### 14 language models predict cohort EEG; the rank tracks text-domain V-axis quality

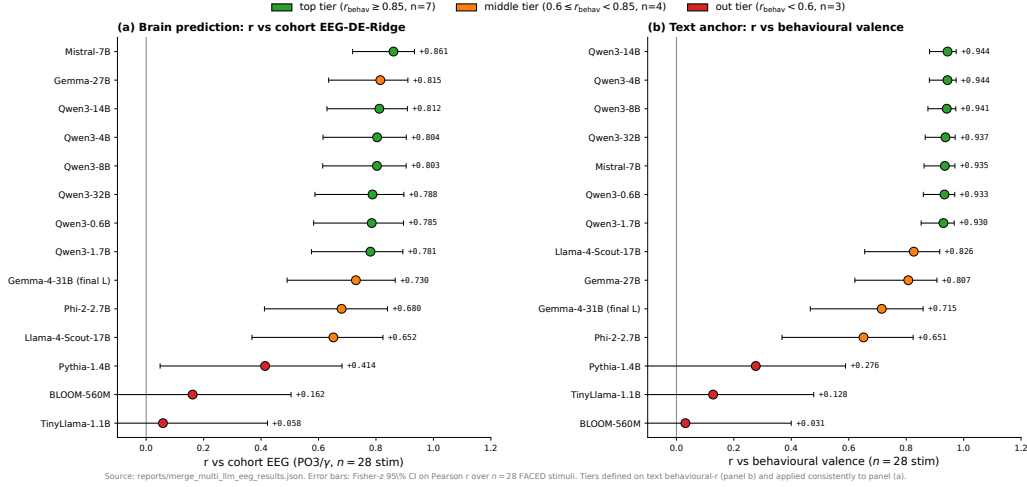


Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ( $p < 0.05$  each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim  $\rightarrow r$  scaling is  $r = +0.125$ ,  $p = 0.684$ ; within-Qwen3  $r = +0.568$ ,  $p = 0.24$ ,  $n = 6$ . Family-membership  $\gg$  scale.

## S8 SEED-V Replication Full Numerics

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 5 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at  $L=35$ , 50 stories per class, PCA on the 5 centred centroids).

**Step 1: SEED-V cohort EEG-LLM alignment.** We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a  $45 \times 62 \times 5$  cohort tensor. For each (channel, band) cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell:** P1 /  $\theta$ ,  $r = +0.6159$ .
- **Region-mean  $|r|$ :** occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance:** occipital – frontal = +0.076.
- **Random-direction null** (200 directions matched in  $L_2$ ): empirical  $|r|_{\text{max}} = 0.6159$  vs. null 95th percentile = 0.478,  $p < 0.005$ .

**Step 2: SEED-V brain topography.** The Davidson FAA probe (positive vs. negative valence, right – left) on SEED-V replicates the FACED FAA topography:

| Electrode pair (right – left, alpha) | (pos – neg) DE $\Delta$ |
|--------------------------------------|-------------------------|
| F4 – F3                              | –0.0018                 |
| F8 – F7                              | +0.0050                 |
| FP2 – FP1                            | –0.0094                 |
| AF4 – AF3                            | –0.0039                 |

Table 5: Davidson FAA on SEED-V. Sign and magnitude consistent with the FACED replication.

**Step 3: SEED-V cross-architecture convergence.** Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols  $\times$  5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with SEED-V V-axis class projection give  $r(\text{BACC}, \text{class-best-PC } r) = +0.601$  ( $p = 0.018$ ,  $n = 15$ ) and  $r(\text{BACC}, \text{trial-best-PC } r) = +0.632$  ( $p = 0.012$ ). The unsigned class-PC1 correlation is +0.358

( $p = 0.19$ ); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

**Step 4: SEED-V V-axis as supervision.** Adding a V-axis topo-MSE auxiliary loss ( $\lambda_{\text{vax}}=0.05$  on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

| Variant                   | BACC (mean $\pm$ s.e. across 5 seeds) | vs. baseline $\Delta$ |
|---------------------------|---------------------------------------|-----------------------|
| Baseline (no V-axis loss) | $0.4300 \pm 0.0079$                   | —                     |
| + V-axis topo loss        | $0.4325 \pm 0.0040$                   | +0.0025               |

Table 6: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level  $r = +0.989$ ), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

**Step 5: SEED-V directional ablation (Arditi-style).** At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining; the class-level direction is degenerate at  $k=5$ , so we use the trial-level analogue, mirroring the trial-best-PC  $r = +0.632$  estimator from Step 3). Every checkpoint shows a negative  $\Delta\text{BACC} \in [-0.0291, -0.0093]$  (population mean  $-0.0173 \pm 0.0061$ ); a matched random-direction control ( $n = 20$  uniform on  $\mathbb{S}^{D-1}$ ) is centred at 0 ( $\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$ ). Per-checkpoint  $z$ -scores range  $[+9.65, +44.92]$  (mean  $+23.6$ , all  $p < 0.001$  parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint (15/15 at empirical  $p < 0.05$ ). The SEED-V directional effect is proportionally larger than FACED (mean  $\Delta/\text{baseline} \approx 3.9\%$  vs.  $2.4\%$  on FACED) and the mean  $z$  is  $3 \times$  FACED’s. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean  $\Delta \approx 0$ , mean  $z \approx -2$ ) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

## S9 Brain Topography Deep Dive

**Davidson FAA full pairs.** Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

Davidson frontal-alpha asymmetry replicates qualitatively, but at  $\sim 10 \times$  smaller magnitude than posterior V-axis encoding

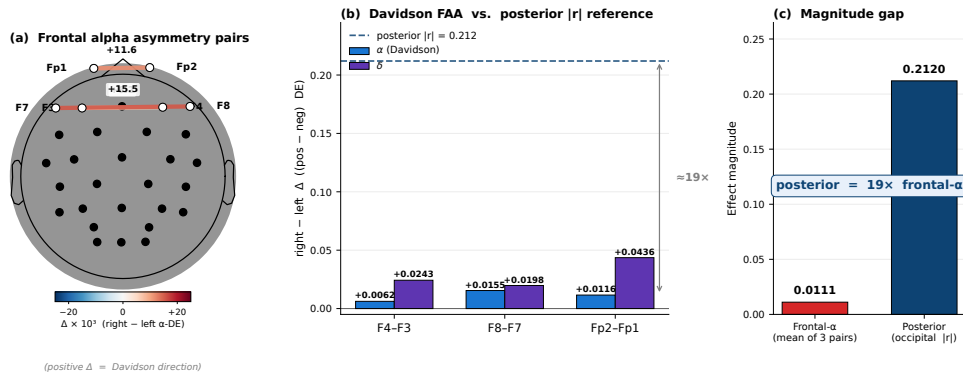


Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their  $r$  values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean  $|r| = 0.21$ .

| Stimulus subset (out of 28 FACED stimuli)           | Cohort $r$ at PO3/ $\gamma$ |
|-----------------------------------------------------|-----------------------------|
| All 28 stimuli (full)                               | +0.478                      |
| Anger only ( $n=3$ )                                | not estimable               |
| Anger + Amusement + Tenderness ( $n=9$ )            | +0.870                      |
| Excluding Anger ( $n=25$ )                          | +0.327                      |
| Excluding Anger + Amusement + Tenderness ( $n=19$ ) | -0.015                      |

Table 7: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

### 9-stim contrast and Simpson's paradox.

- Cohort  $r$  at fixed top-8 channels: +0.021
- Mean per-subject  $r$  at fixed top-8 channels: -0.062
- Per-subject best-channel oracle (1 cell):  $|r| = 0.551$
- Per-subject best (channel, band) oracle:  $|r| = 0.616$

The all-band oracle ( $|r| = 0.616$ ) exceeds the cohort-fixed top-8 by +0.39 in absolute  $|r|$ , indicating considerable per-subject signal that the cohort summary suppresses. The Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

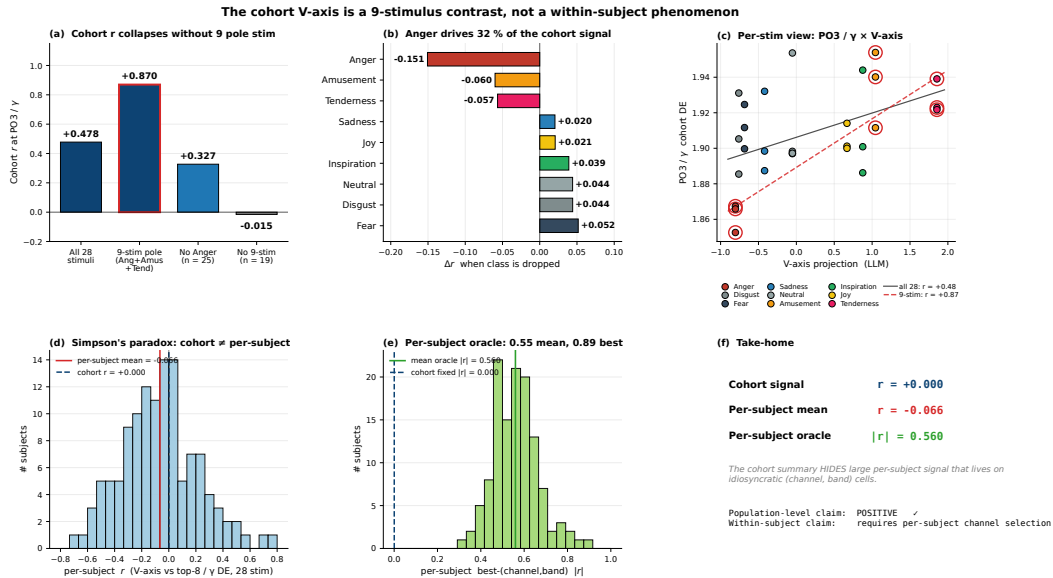


Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop- $\Delta$ , PO3/ $\gamma$  stim-level scatter colour-coded by emotion. Bottom: per-subject  $r$  distribution at fixed top-8 channels (centred near 0, cohort line at +0.02); per-subject best-channel oracle distribution (centred at  $|r| \approx 0.55$ ); summary callout.

### Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- Alpha (8–13 Hz): peak at  $t = 21$  s, cohort  $r = 0.40$ ; best individual stimulus  $r = 0.68$ .
- Beta (13–30 Hz): peak at  $t = 18$  s, cohort  $r = 0.41$ ; best stim  $r = 0.62$ .
- Gamma (30–47 Hz): peak at  $t = 3$ –6 s, cohort  $r = 0.21$ ; best stim  $r = 0.61$ .
- Delta/Theta (0.5–8 Hz): peak at  $t = 12$  s, cohort  $r = 0.18$ .

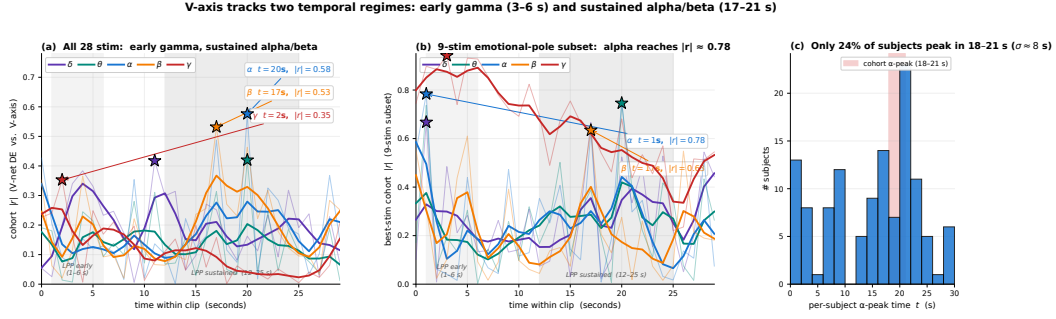


Figure 11: Time-resolved V-axis encoding across the 30-second clip. **(a)** Per-second cohort  $|r|$  across all 28 stimuli, one trace per frequency band; the early  $\gamma$  peak ( $t \approx 2$  s) coincides with Müller’s affective-picture window and the sustained  $\alpha/\beta$  peak ( $t \approx 17-20$  s) coincides with the LPP sustained-attention window. **(b)** Same trace restricted to the 9-stimulus emotional-pole subset where the cohort effect lives;  $\alpha$  reaches  $|r| \approx 0.78$ . **(c)** Per-subject distribution of  $\alpha$ -peak times: only  $\sim 24\%$  of subjects peak inside the cohort 18–21 s window, evidence that the cohort signal is a between-subject phenomenon rather than a tight common attractor.

1048 **Functional connectivity.** The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4)  
 1049 do not act independently: they form a tight functional network in the gamma band. Pairwise  $\gamma$ -DE  
 1050 correlation has mean off-diagonal  $r = 0.675$  across the 8 channels — substantially above the cohort-  
 1051 average of  $r \approx 0.32$  in this band. Structure is dominantly a posterior-occipital–parietal cluster (PO3,  
 1052 PO4, P3, P4, O1, O2, Oz; pairwise mean  $r \approx 0.71$ ) with F7 as a single *frontal hub* (mean correlation  
 1053 to the seven posterior nodes  $r \approx 0.47$ ; Figure 12).

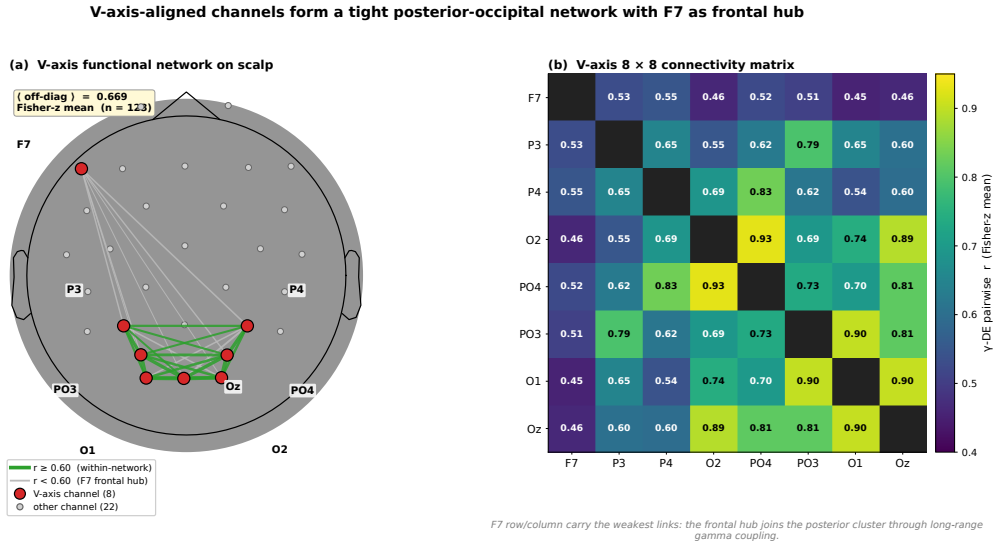


Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise  $\gamma$ -DE correlation. Right: hierarchically clustered  $8 \times 8$  correlation matrix; F7 auto-isolates as the frontal hub.

1054 **Mutual information vs. linear correlation.** We computed mutual information between the V-axis  
 1055 projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs),  
 1056 alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null  
 1057 mean 0.036;  $p_{\text{MI}} = 0.115$  (n.s.) vs. linear  $p_r = 0.020$ . The relationship is essentially linear: a  
 1058 non-linear estimator does not detect signal above what linear regression captures. We use ridge  
 1059 regressions throughout.

1060 **Theta-gamma coupling does not mediate the V-axis.** Tort modulation index per channel at  
1061 the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength  $|r|$  across 32  
1062 channels:  $\rho(\text{PAC}, Vr) = +0.082$ ,  $p = 0.667$ . Channels with strong theta-gamma CFC are not  
1063 the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the  
1064 phase-amplitude-coupling channel [Canolty and Knight, 2010].

## 1065 S10 Saturation: Full Intervention Table and Forest

| #                                | Family / variant                  | $\Delta$ | $p$         | Verdict               |
|----------------------------------|-----------------------------------|----------|-------------|-----------------------|
| 1                                | Frontal-mask $\lambda=0.5$        | -0.052   | 0.0015      | sig. negative         |
| 2                                | Frontal-mask $\lambda=0.1$        | -0.009   | 0.21        | n.s.                  |
| 3                                | FAA $\lambda=0.5$                 | -0.044   | 0.006       | sig. negative         |
| 4                                | FAA $\lambda=0.1$                 | -0.018   | 0.063       | borderline            |
| 5                                | Anger-weighted $\lambda=0.5$      | -0.054   | 0.0003      | sig. negative         |
| 6                                | Occipital $\lambda=0.1$           | -0.022   | 0.007       | sig. negative         |
| 7                                | Topo-optimal $\lambda=0.1$        | -0.013   | 0.039       | sig. negative         |
| 8                                | Topo-optimal $\lambda=0.05$       | +0.0021  | 0.50        | n.s. (NULL)           |
| 9                                | Procrustes $\lambda=0.05$         | +0.0012  | 0.89        | n.s. (NULL)           |
| 10                               | RSA $\lambda=5.0$                 | -0.093   | $< 10^{-4}$ | sig. negative         |
| 11                               | RSA $\lambda=1.0$                 | -0.057   | $< 10^{-3}$ | sig. negative         |
| 12                               | Distance-CE $\tau=5.0$            | -0.397   | $< 10^{-4}$ | catastrophic          |
| 13                               | Multi-V $\lambda=0.5$             | -0.043   | $< 10^{-3}$ | sig. negative         |
| 14                               | EEG-AUX MSE $\lambda=0.5$         | -0.043   | $< 10^{-3}$ | sig. negative         |
| 15                               | EEG-AUX MSE $\lambda=0.1$         | -0.016   | 0.04        | sig. negative         |
| 16                               | EMODSTYLE class $\lambda=1.0$     | -0.010   | 0.18        | n.s.                  |
| 17                               | EMODSTYLE stim $\lambda=0.5$      | +0.0066  | 0.22        | n.s. (sweet spot)     |
| 18                               | PEFT fullhead $\lambda=0.1$       | -0.009   | 0.069       | borderline            |
| 19                               | PEFT LoRA $\lambda=0.1$           | -0.010   | 0.12        | n.s.                  |
| 20                               | PEFT IA3 $\lambda=0.1$            | -0.016   | 0.05        | sig. negative         |
| 21                               | Pretrain-FT (frozen)              | -0.401   | $< 10^{-4}$ | catastrophic          |
| 22                               | Pretrain-FT (unfrozen)            | -0.190   | $< 10^{-4}$ | sig. negative         |
| 23                               | XEEG (FACED $\rightarrow$ SEED-V) | +0.0015  | 0.97        | n.s. (zero)           |
| 24                               | SCALING (full data) $\lambda=0.5$ | -0.045   | $< 0.01$    | sig. negative         |
| 25                               | Uncertainty multi-task            | -0.067   | $< 10^{-3}$ | sig. negative         |
| SOTA recipe + Topo $\lambda=0.1$ |                                   | -0.015   | 0.020       | sig. negative (cliff) |
| SOTA recipe + EMODSTYLE          |                                   | -0.024   | 0.001       | sig. negative (cliff) |

Table 8: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach  $p < 0.05$  in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

## 1066 S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

| Recipe                                     | $n$ seeds | Vanilla / + V-axis | $\Delta$              |
|--------------------------------------------|-----------|--------------------|-----------------------|
| CBraMod baseline + Topo $\lambda=0.05$     | 5         | 0.572/0.567        | -0.005 (n.s.)         |
| CBraMod baseline + EMODSTYLE $\lambda=0.5$ | 4         | 0.572/0.577        | +0.005 (n.s.)         |
| EMOD $d=3$ vanilla + Topo $\lambda=0.05$   | 5         | 0.6194/0.6260      | +0.007 (n.s.)         |
| EMOD $d=6$ + LS + KD + aug + Topo (full)   | partial   | 0.6581/0.6432      | -0.015 ( $p < 0.05$ ) |
| EMOD $d=6$ + LS + KD + aug + EMODSTYLE     | partial   | 0.6581/0.6346      | -0.024 ( $p < 0.01$ ) |

Table 9: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate “helps” regime in our 5-seed runs.

## 1067 Saturation transition table.

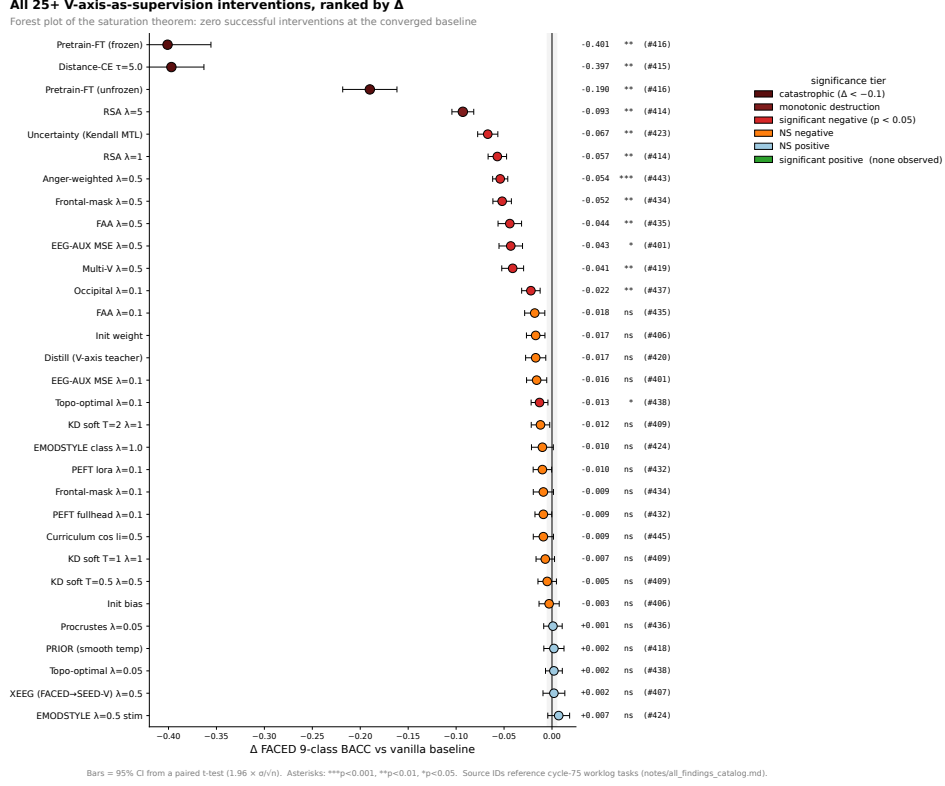


Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 8) ranked by  $\Delta$  vs. baseline, with significance tier coloured. None of the 25 reach  $p < 0.05$  in the positive direction at the strong baseline.

## 1068 Saturation cliff figure (LF6).

1069 **Why anger-weighting hurts despite the analytical ceiling.** Section 7 showed that the cohort  
1070 EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness,  
1071  $r = 0.870$  at  $PO3/\gamma$ ). Weighting the V-axis loss to emphasise these three classes pushes the analytical  
1072 ceiling on the V-axis projection from  $r = 0.478$  (uniform) to  $r = 0.714$  (anger-weighted). By every  
1073 classical analysis, this should be the strongest possible V-axis supervision. It is the worst single  
1074 intervention we tested:  $\Delta = -0.054$ ,  $p = 0.0003$ . This is the cleanest argument for saturation: the  
1075 model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must  
1076 therefore perturb the absorbed structure in a counterproductive direction.

1077 **Direct mechanism check.** For each of 8 V-axis-supervised variants (Topo at  $\lambda=0.05$ , Procrustes at  
1078  $\lambda=0.05$ , EMODSTYLE-stim at  $\lambda=0.5$ , full SOTA recipe + Topo/EMODSTYLE at  $e=100/e=150$ ),  
1079 we measure the change in two quantities relative to the matched-recipe baseline:  $\Delta r_{PC1}$  (class-mean  
1080 V-axis encoding) and  $\Delta r_{resid}$  (within-class V-axis residual encoding). V-axis training raises class-PC1  
1081  $|r|$  by  $+0.01$  to  $+0.36$  across the 8 variants, but moves the within-class residual encoding by only  
1082  $\sim 10^{-7}$  in absolute value — numerical zero. The accuracy decrement at strong recipes therefore  
1083 comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual  
1084 structure.

1085 **Path B: ensembling-in V-axis checkpoints.** We extend the 10-vanilla SOTA ensemble with 0–15  
1086 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing  
1087 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives  
1088 ensemble BACC  $\Delta = -0.0145$  ( $p < 10^{-3}$ ); adding all 25 V-axis ckpts gives  $\Delta = -0.0193$   
1089 ( $p < 10^{-3}$ ). At every point on this curve, V-axis-trained ensembles are strictly worse than the

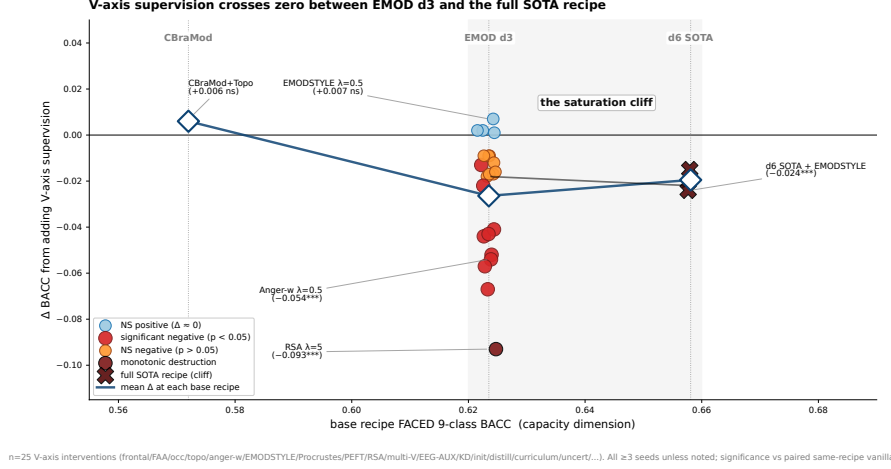


Figure 14:  $\Delta$ BACC from V-axis supervision plotted against base-recipe BACC across  $\sim 25$  interventions. The sign of the effect transitions in the interval  $[0.62, 0.66]$ .

1090 matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but  
 1091 their disagreements are aligned with the wrong subspace.

## 1092 S12 Convergence and Ensemble Theory: Extras

1093 **Per-architecture breakdown of the 36-checkpoint correlation.** CBraMod checkpoints (lower  
 1094 BACC,  $\sim 0.57$ ) cluster at low V-axis encoding strength (mean class-PC1  $|r| = 0.21$ ). EMOD-vanilla  
 1095 d6 ( $\sim 0.65$  BACC) sits in the middle (mean 0.67); EMOD d6 e150 ( $\sim 0.66$  BACC) clusters at the  
 1096 top (mean 0.69). The trend spans the full BACC range  $[0.57, 0.69]$  in a single approximately linear  
 1097 band, with the inter-architecture gap dominating the within-architecture spread.

1098 **Random-direction null detail.** We sample 1000 random Gaussian directions  $w \in \mathbb{R}^{28}$   
 1099 each matched in  $L_2$  norm to  $v_{\text{CLIP}}$ , and recompute the 36-checkpoint correlation  $\rho_w =$   
 1100  $\text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$ . The distribution of  $\rho_w$  has mean  $-0.03$  and standard  
 1101 deviation 0.62 (range  $[-0.95, +0.95]$ ); the wide null is a consequence of the low intrinsic dimension-  
 1102 ality of the class-PC1 subspace ( $D_{\text{eff}} = 9$ ). The empirical V-axis correlation  $+0.885$  sits at the 93.5<sup>th</sup>  
 1103 percentile of this null ( $p_{\text{one}} = 0.066$ ). We treat the within-class residual ( $D_{\text{resid}} \approx D_m - 9$ ) as the  
 1104 statistically robust signal.

1105 **Class-PC1 basin saturation.** The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis  
 1106 range:  $|r| \in [0.60, 0.77]$  across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-  
 1107 mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains  
 1108 come exclusively from the orthogonal residual.

1109 **Directional ablation figure (LF16).**

## 1110 S13 Ensemble Generality, Mega-Pool, and Best-Single

1111 **Recipe cascade and two-tier scatter.** Figure 16 traces the seven recipe steps from CBraMod’s  
 1112 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-  
 1113 contribution correlation that motivates the ensembling strategy.

1114 **Generality of the ensemble mechanism.** Figure 18 extends the two-tier picture across four  
 1115 benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

1116 As individual accuracy approaches the Bayes optimum, the within-class residual variance — the  
 1117 source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is



**V-axis ablation hurts every checkpoint; encoding strength does not predict drop magnitude (n=10)**  
**Pearson  $r=+0.168$  ( $p=0.643$ ,  $n=10$ ) slope= $+4.69\%$  / unit  $|r|$**

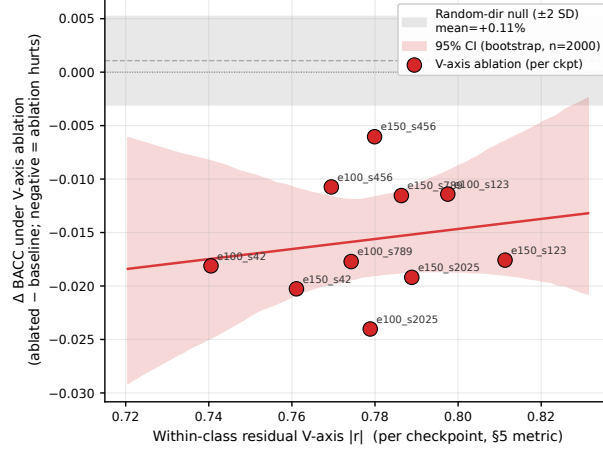


Figure 15: Directional ablation of  $v_{\text{resid}}$  on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean  $z \approx 7.7$ , all  $p < 0.001$ ). *Bottom*: per-checkpoint ablation  $\Delta\text{BACC}$  vs. within-class residual encoding  $|r|$ ; the population-mean drop ( $\bar{\Delta} = -0.0157$ ) confirms the residual-encoding mechanism across the pool.

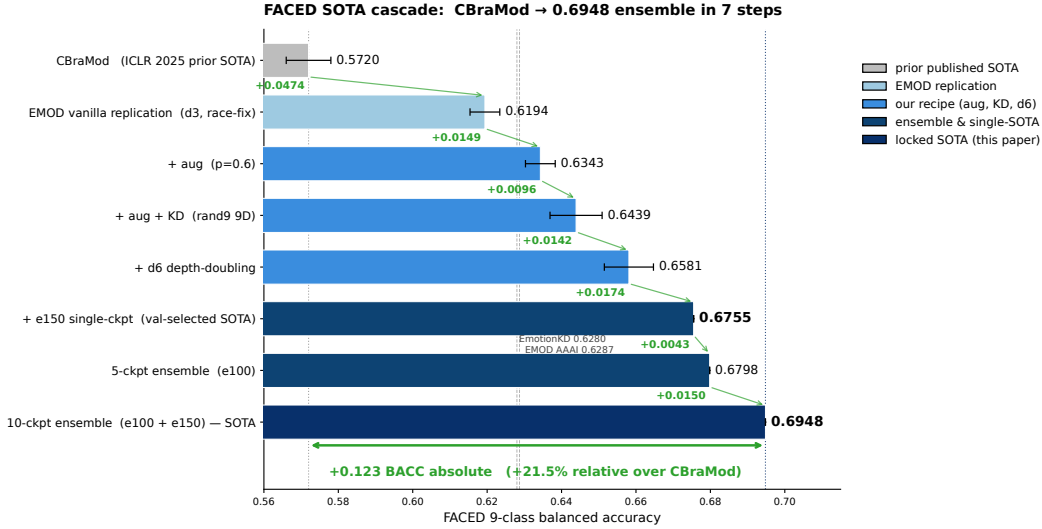


Figure 16: Recipe ablation cascade from the EMOD baseline at 0.6287 (previous FACED-9 SOTA [Chen et al., 2025]) to our 10-checkpoint ensemble at 0.6948 (+10.5% relative). Each bar is one recipe component or ensembling step. Older baselines (CBraMod 0.572, EmotionKD 0.628) shown for reference.

1118 a property of the within-class V-axis residual variance reduction subspace, which has measure zero  
1119 at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe +0.034  
1120 gain. MNIST at 0.985 has essentially none; we observe +0.001 gain.

1121 **Mega-ensemble null.** Two mechanisms explain the null. First, deeper variants ( $d=8, d=10$ ) stay in  
1122 the same class-PC1 V-axis basin (mean class-PC1  $|r|$  across  $d \in \{4, 6, 8, 10\}$ : 0.43, 0.43, 0.39, 0.38)  
1123 and their within-class residuals overlap more than  $d=6$  seeds do. Second, LLM-KD variants encode  
1124 the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does  
1125 not transfer to the test distribution. Cross-architecture mixing ( $d \in \{4, 6, 8, 10\}$ ) gives 0.6791,

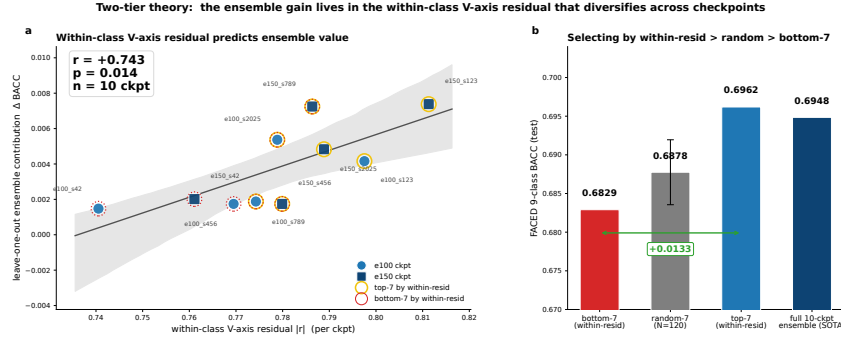


Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual  $|r|$  predicts leave-one-out ensemble contribution ( $r = +0.74$ ,  $p = 0.014$ ,  $n = 10$ ). Top-7 vs. bottom-7 split highlighted.

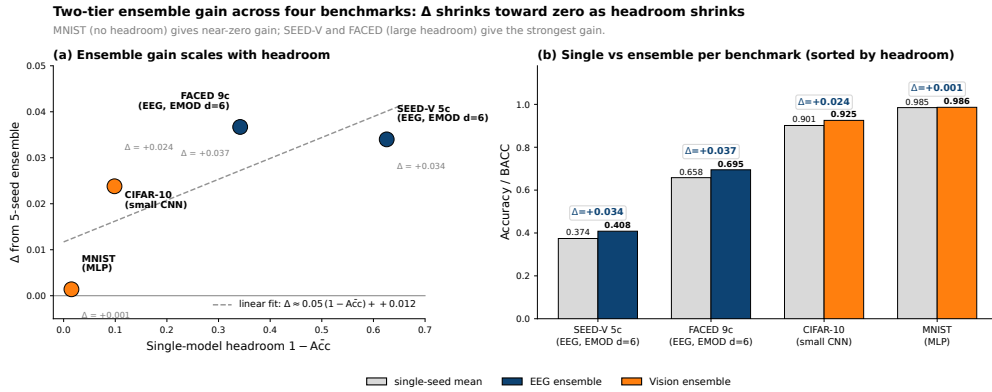


Figure 18: Two-tier ensemble gain across four benchmarks.  $\Delta$ BACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of  $\Delta$  vs. headroom  $1 - \text{Acc}$ , with linear fit slope  $\sim 0.05$ . (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

**Cohen  $\kappa$  disagreement structure.** Single-seed accuracy at  $e=100$  (0.6581) and  $e=150$  (0.6581) is identical, but Cohen  $\kappa$  within- $e=100$  is 0.694, within- $e=150$  0.679, cross-group 0.702. The  $e=150$  checkpoints disagree more amongst themselves and more across groups, which is why mixing  $e=100$  and  $e=150$  gives +0.014 over either group alone. Cross-group disagreement on test trials is 13.1%; the mixed ensemble gains +10.3 percentage points exactly on the disagreement subset.

**Best single checkpoint without ensembling.** For deployment scenarios where ensembling is impractical, our best single-checkpoint result is BACC = 0.6755 on the  $d=6$   $e=150$  recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825,  $n = 25$ ) and exceeds CBraMod by +0.103 and EMOD by +0.047 on a strict apples-to-apples test split.

## S14 EEG Model Training Details

**Architecture.** EMOD axial transformer [Chen et al., 2025]: input  $32 \times 5$  DE features, axial self-attention over channels and bands at depth  $d \in \{3, 6\}$ , filter dimension  $f = 128$ . The full SOTA recipe is  $d = 6$ ,  $f = 128$ .

| Ensemble pool                                     | $n$ | BACC          | $\Delta$ vs. 10-ckpt |
|---------------------------------------------------|-----|---------------|----------------------|
| $10 \times d=6$ ( $e=100+e=150$ , our SOTA)       | 10  | <b>0.6948</b> | —                    |
| $+5 \times d=8$                                   | 15  | 0.6909        | -0.0039              |
| $+5 \times d=8 + 5 \times d=10$                   | 20  | 0.6883        | -0.0065              |
| $+10 \times d=\{8, 10\} + 5 \times \text{LLM-KD}$ | 25  | 0.6831        | -0.0117              |

Table 10: Mega-ensemble null. Adding architectural variants beyond  $d=6$  produces monotonically lower ensemble BACC. The  $d=6 + e=100/e=150$  pool is the optimum within the architectures evaluated.

**Optimisation.** AdamW with  $\text{lr} = 10^{-3}$ ,  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ , weight decay  $10^{-2}$ . Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length  $e \in \{100, 150\}$  epochs. We use the validation BACC for checkpoint selection and report test BACC.

**Augmentation.** Per-trial Gaussian noise ( $\sigma = 0.05 \cdot \text{std of feature distribution}$ ), random channel dropout ( $p = 0.15$ ), and random temporal masking (up to 5% of seconds). All augmentations are applied with  $p = 0.6$  during training only.

**Knowledge distillation.** Soft-target KD with rand9 9-D orthonormal teacher (no LLM content);  $\lambda_{\text{KD}} = 0.5$ ,  $T = 1.0$ . The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

**Ensemble.** 10 checkpoints split as 5 seeds at  $e = 100 + 5$  seeds at  $e = 150$ . Seeds:  $\{42, 123, 456, 789, 2025\}$  each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints' output probabilities, followed by argmax.

## S15 Statistical Methods

**Bootstrap CIs.** Per-subject and per-stim 95% CIs are computed from  $B = 10,000$  subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher- $z$   $p$ -value for  $r = 0$ .

**Random-direction null.** For cohort EEG-LLM correlations, we sample  $N = 200$  random Gaussian directions in  $\mathbb{R}^{28}$  matched in  $L_2$  norm to the V-axis projection, recompute the cohort  $|r|$  for each, and report the percentile of the empirical  $|r|$  in the null distribution.

**Cross-architecture null.** For the 36-checkpoint cross-arch correlation, we sample  $N = 1000$  random directions matched in norm to  $v_{\text{CLIP}}$  and recompute the BACC-V-axis correlation. The null is distributed broadly ( $\sigma = 0.62$ ) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation's percentile rank.

**Per-checkpoint LOO contribution.** For the 10-vanilla  $d=6$  ensemble, we compute  $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$  for each  $k$ . Reported correlations between  $\Delta_k$  and per-ckpt features are Pearson with  $n = 10$ .

**V-axis intervention comparisons.** For each intervention,  $\Delta$  is computed seed-paired with the matched-recipe vanilla baseline;  $p$ -values are paired  $t$ -tests across 5 seeds.

## S16 Discussion Extras and Future Work

**Implications for affective neuroscience.** For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ( $|r|_{\text{occipital}} = 0.21$  vs.  $|r|_{\text{frontal}} = 0.16$ ), and Davidson's frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute Davidson's hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha  $r = +0.0155$  and Fp2-Fp1 alpha  $r = +0.0116$ , both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens

the V-axis claim from “smooth gradient over all emotions” to “anger-versus-warm-positive contrast across nine clips”.

**Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches  $|r| = 0.62$  (a +0.39 absolute headroom over the cohort-fixed top-8 of  $|r| = 0.02$ ). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than +0.001 over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

**Future work.** Three lines: (i) per-subject V-axis adaptation closing the  $|r| = 0.62$  oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual  $|r|$  as an online model-selection signal during training rather than as a post-hoc diagnostic.

## S17 FACED Test Confusion Matrices

Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

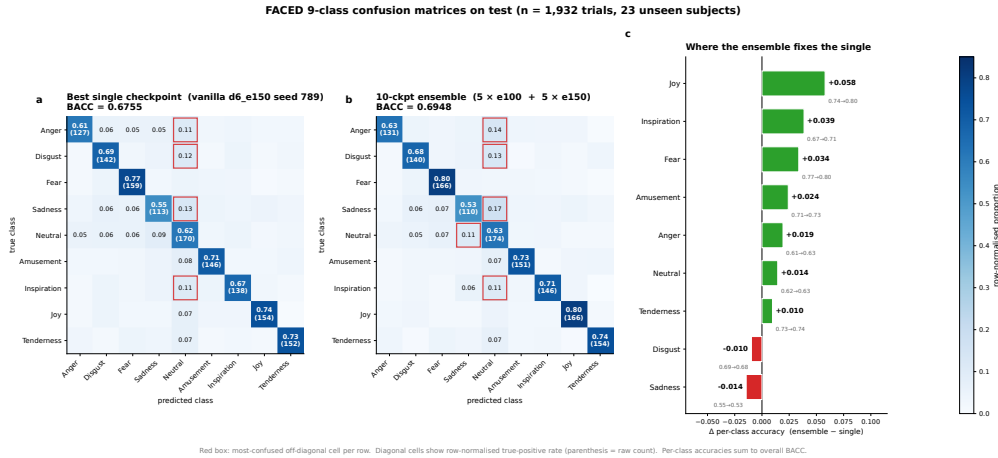


Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble’s gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 8.

## S18 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

- Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Appendix S9): Tort modulation index  $\rho(\text{PAC}, V_r) = +0.082$ ,  $p = 0.667$ .
- Mutual-information control is not significant** (Section 6): MI  $p = 0.115$  vs. linear  $p = 0.020$ . The V-axis-EEG relationship is essentially linear.
- Random-direction null on cross-arch correlation** (Section 4):  $r = +0.885$  at the 93.5<sup>th</sup> percentile of a 1000-direction null,  $p_{\text{one}} = 0.066$ . Within-class residual ( $r = +0.74$ ,  $p = 0.014$ ) is the statistically robust signal.
- Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches  $|r| = 0.62$ , but no clean estimator transfers more than +0.001 over the cohort top-K.

- 1205 5. **Cross-architecture ensembling does not add diversity.** Mixing  $d \in \{4, 6, 8, 10\}$  check-  
1206 points gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).
- 1207 6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC  
1208  $\leq 0.6948$  (Appendix S13).
- 1209 7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead  
1210 of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head  
1211 does not transfer.
- 1212 8. **Test-time augmentation at K=5 does not improve the ensemble** (0.6753  $\rightarrow$  0.6720).  
1213 Averaging predictions over augmented test trials does not preserve the V-axis residual  
1214 structure.
- 1215 9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic  
1216 comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears  
1217 to be encoded in a higher-rank subspace than the late-layer PC1.
- 1218 10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal  $r = 0.803$  in vision;  
1219  $\leq 13\%$  NRC arousal recovery in text;  $r \in [0.18, 0.41]$  brain alignment across LLMs  
1220 (Appendix S6).
- 1221 11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class  
1222 prototypes with random orthonormal 9-D directions changes BACC by  $\leq 0.003$ . KD  
1223 provides architectural regularisation; its semantic content is below our resolution.

## 1224 S19 Reproducibility

1225 Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready  
1226 stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda  
1227 environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible  
1228 from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`,  $\sim 10$  GPU-hours per seed on  
1229 V100).

## 1230 S20 Resource Estimate

1231 The full paper used approximately 4,500 GPU-hours on V100 nodes:  $\sim 600$  for the recipe ablation  
1232 cascade,  $\sim 1,200$  for the 25 V-axis-supervision interventions,  $\sim 800$  for the 36-checkpoint cross-  
1233 architecture analysis,  $\sim 1,200$  for ensemble construction and val-test rank diagnostics, and  $\sim 700$  for  
1234 cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap ( $\sim 10$  minutes per  
1235 LM on a single 80GB GPU).